

# Lead-biased Grounding Score: injecting discourse-structural knowledge into reference-free, cost-constrained assessment of summary quality

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## Abstract

Assessing the quality of machine-generated summaries is a decision problem that deployed systems must solve continuously: which model version to release, which candidate to serve, and when to block a regression. The metrics that correlate best with human judgment are unusable in that setting, because they require either a human-written reference summary or a large language-model judge call per candidate. We introduce the Lead-biased Grounding Score, a reference-free evaluation operator that injects one explicit piece of domain knowledge — the journalistic convention that salient content concentrates in the opening sentences of a news article — as a parametric prior over source-sentence positions, and combines it with sentence-level grounding in a small sentence-embedding space. The operator has a single interpretable hyperparameter and needs one embedder forward pass per sentence: no reference, no judge, no fine-tuned encoder. We evaluate fourteen

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metrics on the SummEval benchmark under one controlled hardware budget, with wall-clock runtime measured for every metric. Our operator attains summary-level Spearman correlations of 0.396, 0.227, 0.156 and 0.356 on coherence, consistency, fluency and relevance, and is one of only five Pareto-optimal metrics in the resulting cost-quality plane; it strictly dominates six widely used metrics, including BERTScore, whose mean correlation it exceeds while running ten times faster. Re-running the entire selection procedure inside a cluster bootstrap further shows that the knowledge prior is worth switching on — the no-prior setting is selected in under 1% of resamples — but that its exponent is not identifiable at benchmark scale: the reported value is selected in 18% of resamples, and an unchanged backbone scored on two draws of the same 50 articles agrees with itself on the exponent only 44% of the time. We therefore present the prior as a design element carrying an obligatory, and demonstrably underdetermined, calibration step.

*Keywords:* Summarization quality assessment, reference-free evaluation, knowledge-informed prior, sentence embeddings, cost-aware decision support, hyperparameter calibration

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## 1. Introduction

Every organisation that operates a summarisation service has to answer the same question many times a day: *is this summary good enough?* The question arises when a new model checkpoint is proposed for release, when several decoding configurations compete for the same request, when a nightly regression suite must decide whether a change degraded output quality, and when a reinforcement-learning loop needs a scalar reward for a freshly sam-

pled candidate. In all of these situations the answer is consumed by an automated decision procedure, not by a human reader, and it must be produced quickly, deterministically, and without any artefact that a production pipeline does not have.

The evaluation metrics that agree best with human judgment fail exactly this test. Reference-based metrics, including BLEU [1], ROUGE-L [2], METEOR [3], ChrF [4], BERTScore [5], MoverScore [6], BARTScore [7] and SMART [8], all presuppose a human-written reference summary — a precondition that production inputs never satisfy, because the candidate is the only summary in the system. The two strongest *reference-free* contenders on the SUMMEVAL [9] benchmark avoid that dependency but replace it with an inference cost that dominates the engineering budget: UniEval requires a fine-tuned T5-large pass per (article, candidate, dimension) tuple [10], and G-Eval queries a large language-model judge with a chain-of-thought prompt and aggregates over multiple sampled generations [11]. Recent surveys of reference-free text-generation metrics [12] and of the LLM-as-a-judge paradigm [13] document the same tension: judge-based evaluation buys correlation with human raters at the price of latency, monetary cost, non-determinism and reproducibility.

There is, however, a resource that deployed summarisation pipelines do possess and that current metrics leave unused: *explicit knowledge about how source documents are structured*. News writing follows the inverted-pyramid convention, in which the most newsworthy material is placed in the opening sentences; the resulting *lead bias* of news corpora is one of the best-documented regularities in the summarisation literature [14, 15]. This knowl-

edge is free at inference time — it requires no model, no annotation and no retrieval — and it is exactly the kind of structural domain regularity that a knowledge-informed system should encode explicitly rather than hope a learned representation will rediscover.

### 1.1. Contributions

This paper makes the following contributions.

1. **A knowledge-informed evaluation operator.** We define the Lead-biased Grounding Score (LGS), which combines sentence-level grounding of the candidate in the source with an explicit, one-parameter prior over source-sentence position that encodes discourse-structural knowledge about news documents (Section 3). The operator is reference-free, interpretable — every candidate sentence is anchored to one identifiable source sentence — and runs at one small-embedder forward pass per sentence, with no judge call and no fine-tuned encoder.
2. **A cost-controlled benchmark of fourteen metrics.** We reproduce thirteen published metrics plus LGS on SUMMEVAL under one hardware budget and one runtime harness, and report wall-clock cost per sample alongside correlation for every metric (Section 6). To our knowledge, no prior meta-evaluation of summarisation metrics reports measured deployment cost and correlation on the same axis for a baseline pool of this size.
3. **A Pareto analysis of the cost–quality plane.** The controlled measurement makes it possible to ask which metrics are *not* dominated. Only five of fourteen are Pareto-optimal, and LGS is one of them (Section 6.3). LGS strictly dominates six widely deployed metrics, among

them BERTScore, which it exceeds in mean summary-level Spearman  $\rho$  while running roughly ten times faster. The same analysis shows that the LLM-judge metric in our reproduction is itself dominated by the fine-tuned-encoder metric.

4. **A measurement of how much of a tuned structural prior is signal and how much is the luck of the development draw.**

Metric papers report the selected hyperparameter as a number; we instead re-run the *entire selection procedure* on every cluster-bootstrap resample and report the resulting distribution over grid values (Section 7.3). Two facts follow. Switching the prior on is robust:  $\lambda=0$  is selected in 0.7% of development resamples and 0.3% of test resamples. Which exponent to switch it to is not: the reported  $\lambda^*=0.5$  is selected in 17.8% of resamples, and a fixed backbone compared against itself on two independent draws of the same 50 articles agrees on the exponent only 43.5% of the time. We therefore present LGS as a design plus an obligatory calibration protocol (Section 4) whose output is a usable interval rather than a reproducible point, and we report the width of that interval instead of suppressing it.

5. **A reproducible artefact.** The full pipeline — all fourteen metrics, the calibration sweep, the backbone ablation and every table and figure in this paper — is released as open-source code with a single-command reproduction path at <https://github.com/mikolajsemeniuk/llmbench>.

We are explicit from the outset about what is *not* claimed. LGS does not attain the highest correlation with human judgment in our pool: UniEval

does, and we report that plainly. The claim is that LGS occupies a point in the cost–quality plane that no other metric dominates, and that this point is the operationally relevant one for the automated decision procedures described above.

### *1.2. Structure of the paper*

Section 2 surveys reference-based and reference-free metrics, organises them by the knowledge source they consume, and states the research gap. Section 3 defines LGS, gives its algorithm and computational complexity, and motivates each design choice. Section 4 presents the calibration protocol for the knowledge prior and explains why it is a necessary rather than a decorative step. Section 5 describes the experimental protocol. Section 6 reports the cost–quality analysis, including the Pareto frontier. Section 7 reports correlations with human judgment together with paired-bootstrap significance tests, the prior sweep, the backbone ablation, and the cross-backbone calibration result. Section 8 discusses deployment, limitations and threats to validity, and Section 9 concludes.

## **2. Related Work and Research Gap**

### *2.1. Reference-based metrics*

The longest-standing family of summarisation metrics is reference-based: a candidate summary is compared against one or more human-written reference summaries. N-gram precision and recall metrics (BLEU [1], ROUGE-L [2], METEOR [3], ChrF [4]) operate on surface-form overlap. Their well-documented blind spot is that paraphrases of equivalent content score low

simply because the surface forms differ, which is precisely the regime abstractive summarisation operates in. Contextual-embedding metrics replace surface overlap with cosine similarity in a learned representation space derived from Transformer encoders [16, 17]: BERTScore [5] matches token embeddings greedily, MoverScore [6] solves a transportation problem over them, BARTScore [7] re-uses the encoder–decoder log-likelihood under a pretrained generation model based on BART [18], and SMART [8] extends the matching to sentence units. A parallel line of work in machine translation evaluation learns metric weights directly against human judgments (BLEURT [19], COMET [20]), and on the summarisation side a related family targets faithfulness specifically (FactCC [21] for factual consistency classification, SummaC [22] for NLI-based inconsistency detection). They alleviate the paraphrase problem but inherit the same structural dependence on the reference: the score is a function of (reference, candidate) pairs and is undefined when no reference exists.

## 2.2. Reference-free and source-based metrics

A smaller body of work scores candidates against the *source* article rather than against a human reference, removing the deployment dependency on hand-written summaries; a recent survey maps this family across generation tasks [12]. G-Eval [11] prompts a large language model [23] with a chain-of-thought rubric [24] to score each (source, candidate) pair on the relevant dimension. It obtains the strongest published correlation on SUMMEVAL’s consistency dimension among published metrics, but at a substantial cost: one or more LLM-judge calls per (article, candidate, dimension) tuple, with sampled decoding requiring repeated calls per pair to absorb LLM-

judge variance (the original paper averages 20 GPT-4 samples). Surveys of the judge paradigm document the accompanying issues of prompt sensitivity, length bias, non-determinism and dependence on proprietary, frequently-updated endpoints [13], and a growing line of work explicitly targets cheaper substitutes for judge calls in evaluation [25]. The metric is therefore inappropriate for deployment scenarios that require deterministic, low-latency, high-throughput scoring. UniEval [10] is a hybrid: a fine-tuned T5-large encoder [26] is queried with a Boolean question (“Is this summary {coherent, consistent, fluent, relevant}?”) and the score is the posterior probability of “Yes”. UniEval sidesteps the LLM-judge cost but still requires a fine-tuned encoder per dimension and a forward pass per (article, candidate, dimension) tuple. GPTScore [27] re-uses a pretrained language model to compute a per-token log-likelihood of the candidate conditioned on the source, requiring a single forward pass per pair but inheriting the model’s calibration as the score’s calibration.

### *2.3. The closest control: whole-text embedding cosine*

The simplest reference-free embedding baseline is a single cosine similarity between the embedding of the entire source and the embedding of the entire candidate, computed with the same sentence-embedder used by the metric under test. We call this baseline EMBEDSCORER. Two properties make it ideal for like-for-like comparison with LGS. First, it shares the embedder backbone, so any difference between the two metrics is attributable to the architectural choices of LGS — sentence-level grounding via  $\max_i$  on the source side, mean aggregation on the candidate side, and the position prior  $w(i)$  — and not to a stronger or weaker embedding model. Second, both met-



rics are reference-free; the comparison isolates the architectural contribution from the orthogonal question of whether reference-based metrics are stronger than reference-free ones in general. We return to the EMBEDSCORER-vs-LGS gap as the controlled ablation of the knowledge component in Section 7.

#### 2.4. Research gap

Table 1 organises the metric pool by the knowledge sources each metric consumes at inference time. Read this way, a structural gap becomes visible. Every metric in the pool draws on exactly one of three resources: a human reference, a learned generative or discriminative model of quality, or an LLM judge. None consumes *a priori knowledge about the structure of the source document*, even though such knowledge is available at zero inference cost and is one of the most robust regularities in the news domain that the dominant benchmarks are built on.

Prior systems do exploit lead bias, but only *implicitly* and only on the generation side: the lead-3 extractive baseline is a standing strong competitor on CNN/DailyMail, and pretrained encoder–decoder summarisers absorb the same prior through their training distribution [28, 29, 30]. On the *evaluation* side the prior is unused. LGS closes this gap by making the structural prior an explicit, inspectable, single-parameter component of the scoring function — and, because it is explicit, a component that can be ablated, swept, and shown to be necessary or not, which an implicitly-absorbed prior cannot.

A second gap concerns how metric papers are evaluated. Published meta-evaluations report correlation with human judgment; they do not report what it costs to obtain that correlation. Without a cost axis one cannot tell whether a metric is a sensible deployment choice or merely a strong one, and

Table 1: The metric pool organised by the knowledge sources consumed at inference time. *Ref.* denotes a human-written reference summary, *Src.* the input article, *Model* a pretrained or fine-tuned scorer, *Judge* an autoregressive LLM prompted to rate, and *Prior* explicit a priori knowledge about document organisation. Only LGS consumes the last.

| Metric                      | Family                    | Ref. | Src. | Model | Judge | Prior |
|-----------------------------|---------------------------|------|------|-------|-------|-------|
| BLEU, ROUGE-L, ChrF, METEOR | lexical overlap           | ✓    |      |       |       |       |
| SMART-String                | lexical, sentence         | ✓    |      |       |       |       |
| EMBEDSCORER                 | embedding, doc.           | ✓    |      | ✓     |       |       |
| SMART-Model                 | embedding, sent.          | ✓    |      | ✓     |       |       |
| BERTScore                   | embedding, token-to-token | ✓    |      | ✓     |       |       |
| MoverScore                  | embedding, transp.        | ✓    |      | ✓     |       |       |
| BARTScore, GPTScore         | likelihood                | ✓    |      | ✓     |       |       |
| UniEval                     | fine-tuned QA             | ✓    | ✓    | ✓     |       |       |
| G-Eval                      | LLM judge                 |      | ✓    | ✓     | ✓     |       |
| <b>LGS (this work)</b>      | embedding, sent.          |      | ✓    | ✓     |       | ✓     |

one cannot detect metrics that are dominated — slower *and* weaker than an available alternative. Section 6 supplies that axis under a single controlled budget and shows that domination is common in the current pool.

### 3. The Lead-biased Grounding Score

#### 3.1. Definition

Let  $D$  be a source document split into  $n$  sentences  $s_0, \dots, s_{n-1}$  and let  $C$  be a candidate summary split into  $m$  sentences  $c_0, \dots, c_{m-1}$ . Let  $\text{emb}(\cdot)$  be a sentence embedder of the Sentence-BERT family [31] (in the empirical part, we use `nomic-embed-text` [32] served via Ollama, see Section 5). The Lead-biased Grounding Score is defined as

$$\text{LGS}(D, C) = \frac{1}{m} \sum_{j=0}^{m-1} \max_{0 \leq i < n} w(i) \cdot \cos(\text{emb}(c_j), \text{emb}(s_i)), \quad (1)$$

where the source-position prior is

$$w(i) = \exp(-\lambda \cdot i/n), \quad (2)$$

$i = 0$  denotes the lead sentence, and  $\lambda \geq 0$  is the only hyperparameter.  $\lambda = 0$  disables the prior and reduces LGS to position-agnostic mean-of-max recall. Figure 1 illustrates the operator graphically.

#### 3.2. Algorithm and computational complexity

Algorithm 1 states the procedure. Every step is deterministic given the embedder, and the argmax index recorded in line 10 is the interpretability handle: it names the source sentence that the operator considers to be the evidence for each candidate sentence, so a low score can always be traced to the specific summary sentences that failed to find an anchor.

Let  $d$  be the embedding dimension and let  $T$  be the cost of one embedder forward pass. Embedding costs  $\mathcal{O}((n+m)T)$  and dominates in practice; the similarity stage costs  $\mathcal{O}(nmd)$  floating-point operations and the prior costs

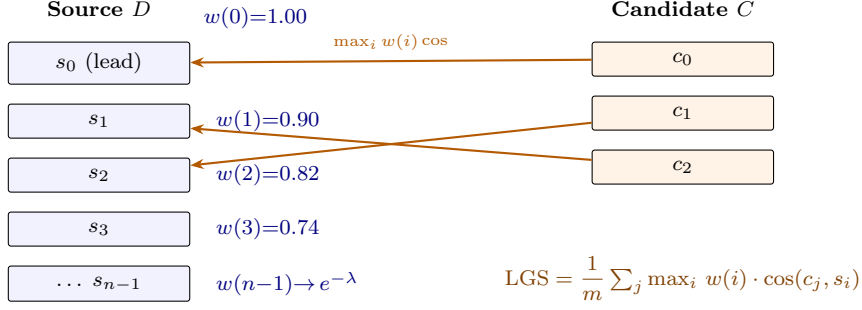


Figure 1: LGS operator. For each candidate sentence  $c_j$ , the metric finds the source sentence  $s_i$  that maximises the position-weighted cosine similarity  $w(i) \cdot \cos(c_j, s_i)$ , where  $w(i) = \exp(-\lambda i/n)$  is an exponential prior favouring the lead. The score is the mean of these per-candidate maxima. Orange arrows indicate illustrative argmax matches.

$\mathcal{O}(n)$ . Memory is  $\mathcal{O}((n+m)d)$ , since the  $n \times m$  similarity matrix need never be materialised — the maximum over  $i$  can be accumulated in a single pass per candidate sentence.

Two properties matter for deployment. First, source embeddings depend only on  $D$ , so when  $K$  candidates are scored against the same article — the standard situation in model selection and in reinforcement-learning rollouts — the source cost is paid once and amortised, giving  $\mathcal{O}((n+Km)T)$  instead of  $\mathcal{O}(K(n+m)T)$ . Second, the exponent  $\lambda$  enters only through the weight vector  $w$ , so a full sweep over a  $\lambda$ -grid of size  $G$  reuses one set of embeddings and costs  $\mathcal{O}(Gnmd)$  additional arithmetic with no additional forward passes. Calibration (Section 4) is therefore essentially free relative to scoring, which is what makes an obligatory per-deployment sweep a realistic requirement rather than a burden.

For contrast, a judge-based metric costs one or more autoregressive decodes per (article, candidate, dimension) tuple, with the decode length itself

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**Algorithm 1** Lead-biased Grounding Score

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**Require:** source document  $D$ , candidate summary  $C$ , exponent  $\lambda \geq 0$ , sentence embedder  $\text{emb}(\cdot)$

**Ensure:** score in  $(-1, 1]$ , plus per-sentence evidence anchors

```
1:  $(s_0, \dots, s_{n-1}) \leftarrow \text{SPLITSENTENCES}(D)$ 
2:  $(c_0, \dots, c_{m-1}) \leftarrow \text{SPLITSENTENCES}(C)$ 
3:  $E_i^s \leftarrow \text{emb}(s_i)$  for  $i = 0, \dots, n-1$   $\triangleright$  cached per source document
4:  $E_j^c \leftarrow \text{emb}(c_j)$  for  $j = 0, \dots, m-1$ 
5: for  $i \leftarrow 0$  to  $n-1$  do
6:    $w_i \leftarrow \exp(-\lambda \cdot i/n)$   $\triangleright$  structural prior, length-normalised
7: end for
8: total  $\leftarrow 0$ 
9: for  $j \leftarrow 0$  to  $m-1$  do
10:   $a_j \leftarrow \arg \max_{0 \leq i < n} w_i \cdot \cos(E_j^c, E_i^s)$   $\triangleright$  evidence anchor for  $c_j$ 
11:  total  $\leftarrow$  total +  $w_{a_j} \cdot \cos(E_j^c, E_{a_j}^s)$ 
12: end for
13: return total/ $m$ ,  $(a_0, \dots, a_{m-1})$ 
```

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dependent on the rubric; a fine-tuned-encoder metric costs one encoder pass per tuple, and thus scales linearly in the number of quality dimensions, which LGS does not.

### 3.3. Worked example

To make the operator concrete, consider a stylised article with  $n=5$  source sentences —  $s_0$ : “Apple announced a new iPhone at its September event.”;  $s_1$ : “The phone features an improved camera system.”;  $s_2$ : “Tim Cook pre-

sented the keynote in Cupertino.”;  $s_3$ : “Pre-orders begin Friday.”;  $s_4$ : “Stock rose 3% after the announcement.” — and a candidate summary with  $m=2$  sentences —  $c_0$ : “Apple unveiled a new iPhone.”;  $c_1$ : “Pricing starts at \$999.”. Suppose, for illustration purposes, that the cosine similarity matrix between the candidate sentence embeddings and the source sentence embeddings is

$$[\cos(c_j, s_i)]_{j,i} = \begin{pmatrix} 0.91 & 0.41 & 0.52 & 0.18 & 0.34 \\ 0.22 & 0.31 & 0.15 & 0.78 & 0.42 \end{pmatrix}.$$

At  $\lambda=0.5$  the position prior takes values  $w(0)\approx 1.00$ ,  $w(1)\approx 0.90$ ,  $w(2)\approx 0.82$ ,  $w(3)\approx 0.74$ ,  $w(4)\approx 0.67$ . For  $c_0$  the position-weighted argmax is  $s_0$  (weighted similarity  $1.00\cdot 0.91=0.91$ ); for  $c_1$  it is  $s_3$  ( $0.74\cdot 0.78=0.58$ ). The score is the mean of these two per-candidate maxima,  $\text{LGS}=(0.91+0.58)/2=0.745$ .

A hallucinated  $c_1$  that is unrelated to the source — say, “Tim Cook resigned as CEO.” with (assumed for illustration purposes) cosine row  $(0.18, 0.09, 0.55, 0.11, 0.21)$  — yields position-weighted maximum  $0.82\cdot 0.55=0.451$  on  $s_2$ , and the score drops to  $(0.91+0.45)/2=0.680$ . The metric’s sensitivity to ungrounded content is structurally tied to the candidate-side mean: with candidate length  $m$ , a single ungrounded sentence removes approximately  $1/m$  of the score’s mass.

### 3.4. Design rationale

*Sentence-level grounding.* A candidate sentence is *grounded* when some source sentence semantically supports it. The LGS metric (Eq. 1) implements this by taking, for each candidate sentence  $c_j$ , the maximum cosine similarity to any source sentence under the position prior, and then aggregating across the candidate with a mean. The asymmetry is

deliberate:  $\max_i$  on the source side is a localised signal (“does this candidate sentence have *any* anchor in the source?”);  $\text{mean}_j$  on the candidate side is a hallucination-sensitive aggregate (a single ungrounded candidate sentence depresses the score by  $1/m$ , where  $m$  is the candidate length, rather than being washed out by an averaging operator on both sides). We compared this asymmetric mean-of-max formulation empirically against alternative aggregations during the calibration phase — low-percentile recall, top- $k$  recall, candidate-side coverage exponent — and report the negative results in Appendix Appendix B; none lifted the development-set mean Spearman over the mean-of-max baseline.

*Which knowledge is injected, and why exponentially.* News summarisation corpora — and CNN/DailyMail in particular [33, 30], the source of all SUMM-EVAL articles — exhibit well-documented *lead bias* [14, 15]: salient information is disproportionately concentrated in the opening sentences of the article, a pattern reinforced by the inverted-pyramid convention of journalistic writing. This is the single piece of a priori domain knowledge that LGS encodes, and it enters the scoring function as a multiplicative source-side prior  $w(i) = \exp(-\lambda i/n)$  rather than being left for the embedder to infer. The exponential family is chosen for three reasons. First, parsimony: it is the simplest one-parameter decay shape that interpolates between a uniform prior ( $\lambda=0$ ) and a near-step prior ( $\lambda \rightarrow \infty$ ) without introducing a discontinuity. Second, scale-invariance: normalising the position  $i$  by the document length  $n$  makes the prior behave consistently across short and long articles, removing the need for a length-dependent tuning of  $\lambda$ . Third, multiplicative interaction: the prior reweights the cosine similarity *before* the  $\max_i$  opera-

tor, so it can move which source sentence wins the argmax for a given candidate; an additive prior would only shift scores uniformly without affecting the matching structure. The choice of an explicit prior over an implicit one is what makes the knowledge component *auditable*:  $\lambda=0$  removes it exactly, so its contribution can be measured rather than asserted (Section 7.2).

*Why a single hyperparameter.* The canonical LGS has exactly one tunable scalar,  $\lambda$ . Possible alternative formulations of this metric can be richer: a precision side mirroring BERTScore’s F-measure structure, a candidate-side coverage exponent that rewarded using more distinct source anchors, a within-summary diversity penalty, and a top- $k$  source aggregation. We tested each of these alternatives on the same held-out development split (Section 4) and rejected them for failing to lift the development-set mean Spearman  $\rho$  over the no-prior baseline. The rejected components and their negative numbers are reported in Appendix Appendix B. The single-hyperparameter design is the one we propose. A beneficial corollary of this discipline is that the calibration protocol has a small, well-defined search space (one scalar over a 1-D grid), which — combined with the embedding reuse noted in Section 3.2 — makes recalibration cheap enough to be mandatory.

*Relation to BERTScore.* The closest already-existing metric is BERTScore [5], which also computes a mean-of-max over cosine similarities between embedded units. However, there are three essential operational differences between LGS and BERTScore. First, BERTScore matches *tokens* (BPE or wordpiece) under a contextual encoder. In contrast, LGS matches *sentences* under a sentence encoder. The unit choice changes the semantic granularity: a token-level  $\max_i$  rewards lexical or stylistic overlap, while a sentence-level



$\max_i$  rewards proposition-level grounding and is structurally less sensitive to surface paraphrase. Second, BERTScore is reference-based: one side of every max is a human reference summary. LGS is reference-free: one side is the *source article*, removing the deployment dependency on human-written summaries (Section 2). The cost-quality implications differ accordingly: BERTScore’s correlation gain over n-gram metrics buys nothing in deployments where no reference exists. Third, neither BERTScore nor any other published mean-of-max metric uses a source-position prior. The lead-bias prior  $w(i)$  in LGS is the architectural ingredient that distinguishes it from a structurally simpler “sentence-BERTScore against the source” baseline; the EMBEDSCORER-vs-LGS comparison in Section 7 is the controlled experiment that isolates this contribution.

## 4. Calibrating the Knowledge Prior

### 4.1. Why calibration is a first-class step

A structural prior is a statement about the source documents, but it acts inside a learned representation space, and the two interact. Two separate findings make the sweep obligatory rather than decorative.

First, the point estimate of the optimum differs across backbones: with `nomic-embed-text` the development-set optimum is  $\lambda^* = 0.5$ , with `bge-m3` [34] it is  $\lambda^* = 0.25$ , and applying the nomic-tuned value to `bge-m3` *underperforms* the no-prior baseline (Section 7.5).

Second — and this is the finding that actually forces the protocol — the exponent is not identifiable from a 50-article development split at all, on any backbone. Section 7.3 measures this directly by re-running the selection

procedure inside the cluster bootstrap: the selected value carries under a fifth of the selection mass, and an *unchanged* configuration disagrees with itself across two draws more than half the time. A practitioner who copies the published number is therefore not merely risking a backbone mismatch; they are copying one draw’s answer to a question that this much data cannot settle. What transfers is the sweep, which costs no additional embedder passes (Section 3.2), and the interval it returns.

#### 4.2. Article-level split

SUMMEVAL contains 100 source articles, each summarised by 16 systems, for 1,600 (source, candidate) pairs with human ratings on four dimensions (coherence, consistency, fluency, relevance). We split at the *article level*, deterministically by JSONL order: the first 50 articles form the development split, the remaining 50 the test split. The article-level split is critical because the 16 candidate summaries of a single article are not independent observations: a difficult article depresses correlations across all systems jointly. A naïve sample-level split would leak across the development/test boundary.

#### 4.3. Selection criterion and verification

We sweep  $\lambda \in \{0, 0.25, 0.5, 1.0, 2.0\}$  on the development split, score each variant with summary-level Spearman  $\rho$  averaged over the four dimensions, and select the best  $\lambda^*$ . The selected  $\lambda^*$  is then re-evaluated on the test split, which is *not* consulted during selection. The verification criterion is whether  $\lambda^*$  outperforms the no-prior baseline ( $\lambda = 0$ ) on the test split’s mean Spearman  $\rho$  — that is, whether the tuned value still helps on data unseen during selection. We do not require the test split’s argmax to match

$\lambda^*$ ; the selection is over the development split only, and a flat optimum (where multiple  $\lambda$  values give near-equal performance) is operationally desirable.

Reporting  $\lambda^*$  as a bare number would leave open how much of it is reproducible, so the selection procedure is itself submitted to the cluster bootstrap of Section 4.5: on each resample of the development articles the full sweep is recomputed and its argmax recorded, giving a distribution over grid values rather than a single winner. Section 7.3 reports that distribution together with two controls that separate a genuine shift in the optimum from ordinary resampling noise. We recommend that any future variant of this protocol report the same distribution; a selection procedure that cannot state how often it would reselect its own output has not finished reporting its result.

#### *4.4. Choice of correlation coefficient*

We report Spearman  $\rho$  as the primary correlation coefficient and Kendall  $\tau$  as a secondary check; Pearson  $r$  is computed internally for diagnostics, but is not shown in the headline tables. The choice of Spearman is consistent with prior SUMMEVAL meta-evaluation: human ratings on the four dimensions are ordinal Likert scores, and the metric value is a real number whose absolute calibration we do not claim. A rank-based correlation is the appropriate statistic for “does the metric order summaries the way the rater does”, which is the operational question. Pearson would additionally penalise non-linear monotonic relationships between the metric and the human score that are immaterial for ranking, and is therefore over-restrictive for the use case.

#### 4.5. Bootstrap and significance testing

For each correlation, we report a 95% confidence interval estimated by *cluster bootstrap* at the article level. This requires resampling the documents with replacement and computing the correlation on the concatenated set of sample-level scores in the resampled documents, with 1,000 resamples for the headline tables and 5,000 for the finer-grained sweeps in Section 7.2. The article-level clustering is essential: SUMMEVAL’s 16 candidates per article are not independent observations, since they share a source whose intrinsic difficulty depresses or lifts all 16 scores together. A naïve sample-level bootstrap would treat the 1,600 pairs as independent, underestimate the variance, and produce confidence intervals that are systematically too narrow [35]. The cluster bootstrap is the canonical fix and is what SUMMEVAL-era meta-evaluation uses.

For metric-vs-baseline comparisons we use a *paired* bootstrap over articles (5,000 resamples): on each resample we compute both metrics’ Spearman on the same resample, record the delta  $\Delta\rho=\rho_{\text{LGS}}-\rho_{\text{baseline}}$ , and report the mean delta, its 95% confidence interval, and a two-sided  $p$ -value computed as twice the smaller of the proportions of  $\Delta$  at or below zero versus at or above zero. Pairing is critical: the two metrics’ correlations co-move with the difficulty of the resampled article subset, and an unpaired comparison of two separately-bootstrapped confidence intervals would inflate the apparent uncertainty by exactly that shared component. The paired procedure also yields directly interpretable “LGS-significantly-better-than-baseline” verdicts, which is the form the comparison table in Section 7 reports.

*Multiple comparisons.* The comparison table tests one hypothesis family — “does LGS correlate with human judgment differently from this baseline on this dimension?” — once per (baseline, dimension) cell, for  $6 \times 4 = 24$  cells. Reading 24 raw  $p$ -values against a per-test threshold of 0.05 would put the expected number of spurious positives near one even if no real difference existed, so every cell also carries  $q$ , its Benjamini–Hochberg adjusted  $p$ -value over the whole table [36]. Controlling the false discovery rate rather than the family-wise error rate is the appropriate choice here because the cells are numerous and positively dependent: they share the target metric’s score vector and are computed on the same resampled articles. A cell is reported as a positive finding only when  $q < 0.05$  *and* the paired-bootstrap interval for  $\Delta\rho$  excludes zero. Every significant verdict in Table 5 survives the correction; the weakest survivor is the coherence comparison against SMART-Model at  $q = .010$ .

#### 4.6. Final scoring

After verification, the headline correlations reported in Section 7 are computed on the full 100-article SUMMEVAL set with  $\lambda = \lambda^*$ . The development/test split is used *only* for hyperparameter selection; the headline numbers are not the test-split numbers.

## 5. Experimental Setup

### 5.1. Dataset

SUMMEVAL [9] is built on the CNN/DailyMail reading-comprehension corpus [33]: 100 CNN/DailyMail articles  $\times$  16 abstractive summarisation

systems = 1,600 (source, candidate) pairs. Each pair has expert human ratings on coherence (Coh), consistency (Con), fluency (Flu) and relevance (Rel) on a 1–5 scale. We use summary-level correlations as the primary measure (Spearman  $\rho$ , Kendall  $\tau$ ); system-level correlations are reported as a secondary check.

### 5.2. *Embedding backbones*

The canonical run of LGS uses `nomic-embed-text` (137M parameters) [32], served locally via Ollama, and fixed before the backbone ablation was run. We additionally repeat the canonical metric in Section 7.4 with `all-minilm` (23M, derived from [37]), `mxbai-embed-large` (335M), and `bge-m3` (567M) [34] to verify that the metric design is not specific to one backbone.

### 5.3. *Baselines and the controlled-budget protocol*

We compare against thirteen baselines spanning the reference-based, reference-free, and mixed families. The *reference-based* family contains BLEU [1], ROUGE-L [2], ChrF [4], METEOR [3], SMART-String, SMART-Model (both [8]), EMBEDSCORER (our whole-text cosine baseline on the same embedder as LGS, included as the like-for-like control), BERTScore [5], MoverScore [6], BARTScore [7], and GPTScore [27]. The *source-based* family contains G-Eval [11]. The *mixed* family — metrics that consume both the source article and a single reference summary — contains UniEval [10].

Because the object of study is the cost–quality trade-off, all fourteen metrics are run under one protocol: the same machine, the same single-threaded runtime harness, the same 1,600 pairs, and the same wall-clock

instrumentation. Enforcing a single budget has a consequence that must be stated plainly: the model-based baselines run in configurations smaller than those of the originating papers. Table 2 lists every deviation and the expected direction of its effect on that baseline’s correlation. This is a deliberate trade — a like-for-like cost axis in exchange for absolute correlation heights that are not directly comparable with published figures — and it is the reason the comparisons in this paper are internal to our reproduction.

These deviations shift the absolute correlation heights of the affected baselines slightly downward, but they do not invert the ordering of metric families on SUMMEVAL: UniEval and G-Eval remain the strongest reference-free contenders even under the budget configurations, and the LLM-judge metric stays distinguishable from the embedding-only metrics by mean correlation. All claims concerning LGS are made against *these reproductions*, not against the original-paper numbers; we state this explicitly so that a reader comparing tables across papers can follow the argument. Minor differences between values reported in the original papers and our computed values are expected in any case, given different hardware and the nondeterminism present in several of the methods.

#### 5.4. Significance testing

For each (LGS, baseline, dimension) cell we run a paired bootstrap over articles (5,000 resamples) on summary-level Spearman, and report the mean delta, the 95% confidence interval, and a two-sided  $p$ -value. The full procedure is described in Section 4.5.

Table 2: Per-baseline deviations from the originally published configurations, imposed by the single-machine budget of the controlled protocol. “Impact” summarises the expected direction on the absolute correlation height of the affected baseline; the qualitative ordering of metric *families* on SUMMEVAL is unchanged in every case.

| Metric     | Original configuration                                                           | Our reproduction                                                                                                                        | Impact                                                         |
|------------|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| MoverScore | Corpus-specific IDF weights over tokens; ELMo / BERT contextual embeddings.      | Uniform IDF; otherwise unchanged.                                                                                                       | Slight ↓ (isolates transport cost from IDF re-weighting).      |
| GPTScore   | GPT-3 (175B) as the conditional log-likelihood scorer.                           | GPT-2 large (~774M) as the conditional log-likelihood scorer.                                                                           | Moderate ↓ (smaller LM, weaker calibration).                   |
| G-Eval     | GPT-4, chain-of-thought rubric, $N=20$ sampled judgments per pair.               | qwen2.5:7b-instruct-q4_K_M [38] via Ollama [39], $N=3$ runs at $T=0.7$ , across-run mean used as canonical score and mean±std reported. | Slight-to-moderate ↓ (smaller judge LLM; budget reproduction). |
| UniEval    | Fine-tuned T5-large per dimension, Boolean query, optional verifier-model chain. | Single Boolean query per (article, candidate, dimension); released T5-large checkpoint, no verifier chain.                              | Slight ↓ (drops the verifier-model refinement).                |

## 6. Cost–Quality Analysis

Metric papers report how well a metric agrees with human raters. They do not report what that agreement costs. For a metric that will be called



inside an automated decision loop, the second number is as binding as the first, and without it one cannot detect the situation that turns out to be common in the current pool: a metric that is both slower and weaker than an alternative already on the shelf. This section supplies the missing axis.

### 6.1. Per-sample cost

The deployment cost of LGS is one sentence-embedder forward pass per source sentence and per candidate sentence. With `nomic-embed-text` (137M parameters) served via Ollama on a single Apple-silicon CPU thread and per-document caching of source embeddings, the canonical SUMMEVAL run — 1,600 (source, candidate) pairs spanning approximately  $6.3 \times 10^3$  cached sentence embeddings (1,273 source plus 5,002 candidate) — completes in roughly ninety seconds ( $\sim 57$  ms per sample; Figure 2). By contrast, G-Eval at our reproduction scale (`qwen2.5:7b-instruct-q4_K_M`,  $N=3$  samples per candidate at temperature 0.7, four dimensions evaluated independently) requires approximately 2.1 hours per single-run pass and approximately 6 hours per  $N=3$  multi-run pass on the same hardware. UniEval requires one forward pass of a fine-tuned T5-large [26] per (article, candidate, dimension) tuple, four dimensions  $\times$  1,600 candidates, totalling approximately 12 minutes wall-clock in our reproduction ( $\sim 430$  ms per sample); the pretrained checkpoints are served via the auxiliary Python server in `cmd/modelsrv`, built on top of the HuggingFace `transformers` library [40]. For metrics that score one dimension at a time — G-Eval and UniEval — the per-sample cost reported throughout is the sum over the four dimensions, because a deployed dimensional scorer must invoke the model once per dimension.

### 6.2. The cost–quality plane

Figure 2 plots all fourteen metrics on a single cost–quality plane: the x-axis is the per-sample wall-clock runtime in milliseconds (log scale), and the y-axis is the mean summary-level Spearman  $\rho$  across the four SUMM-EVAL dimensions. Three regions are immediately readable. The bottom-left band contains the n-gram metrics (BLEU, ROUGE-L, METEOR, ChrF, SMART-String): sub-millisecond cost, mean  $\rho$  in the  $[\text{.08}, \text{.19}]$  range. The middle band contains the contextual-embedding reference-based metrics (BERTScore, MoverScore, BARTScore, EmbedScorer, SMART-Model): hundreds of milliseconds to a couple of seconds per sample, mean  $\rho$  in  $[\text{.06}, \text{.25}]$ . The upper band contains the LLM-judge and fine-tuned-encoder metrics (G-Eval, UniEval): hundreds of milliseconds to ten-plus seconds per sample, mean  $\rho$  in  $[\text{.36}, \text{.47}]$ . LGS lands in a distinct location: a per-sample cost 6 to 22 times below the contextual-embedding band ( $\sim 57$  ms, dominated by the sentence embedder rather than tokenisation), and a mean  $\rho$  ( $\sim .28$ ) above every metric in that band.

### 6.3. Pareto analysis

The plane admits a sharper reading than visual inspection. Call a metric *dominated* if some other metric in the pool is at least as fast and at least as well correlated, with at least one of the two strict; the metrics that are not dominated form the Pareto frontier of the pool. Table 3 reports the outcome.

Only five of the fourteen metrics are Pareto-optimal: ROUGE-L, METEOR, ChrF, LGS and UniEval. Three of them are sub-3-millisecond lexical metrics that survive purely on speed and correlate weakly ( $\rho \leq .184$ ); one is UniEval, which anchors the high-correlation end at .470 and 430 ms. LGS

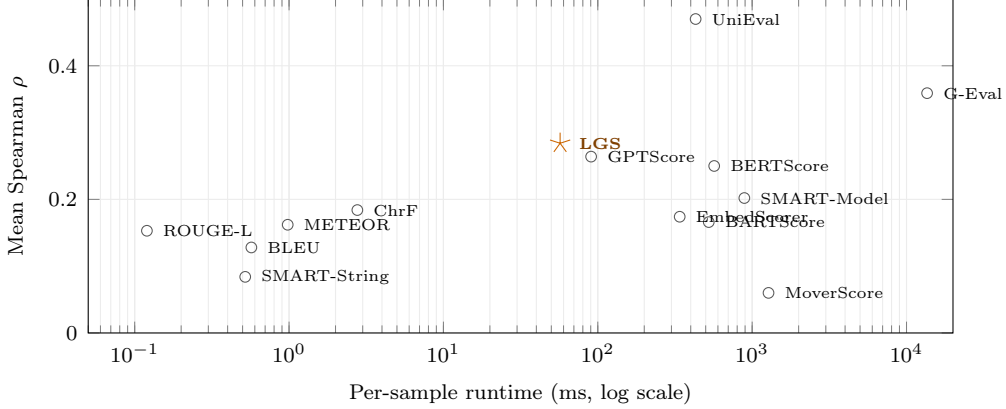


Figure 2: Cost-accuracy frontier on SUMMEVAL. Each point is one metric: x-axis is the per-sample wall-clock runtime under our reproduction (Section 6.1), in milliseconds on a single Apple-silicon CPU thread, on a log scale; y-axis is the mean summary-level Spearman  $\rho$  across the four SUMMEVAL dimensions. LGS (orange star) sits in the lower-cost band of the upper-correlation cluster, distinct from the LLM-judge and fine-tuned-encoder metrics in the upper-right.

is the only metric in between: at 57 ms it is the cheapest metric in the pool with a mean  $\rho$  above .19, and every metric in the pool that is faster correlates at  $\bar{\rho} \leq .184$ .

Three consequences follow.

*LGS strictly dominates six deployed metrics.* It is simultaneously faster and better correlated than EMBEDSCORER (6.0 $\times$ , +.110), GPTScore (1.6 $\times$ , +.020), BARTScore (9.2 $\times$ , +.118), BERTScore (10.0 $\times$ , +.034), SMART-Model (15.6 $\times$ , +.082) and MoverScore (22.4 $\times$ , +.224), where the first figure is the runtime ratio and the second the difference in mean summary-level Spearman  $\rho$ . The BERTScore comparison is the notable one: a 137M-parameter sentence embedder with no access to the human

reference summary attains a higher mean correlation than a RoBERTa-large token-matching metric that does have that access, at one tenth of the runtime. We are careful about the strength of this statement — at the level of individual dimensions the paired bootstrap confirms an advantage only on consistency (Section 7) — but the domination in the (cost, mean  $\rho$ ) plane is unambiguous.

*The LLM judge is dominated in our reproduction.* G-Eval costs  $31.6\times$  more per sample than UniEval and attains a lower mean  $\rho$  (.359 vs .470), so it is off the frontier entirely. This is a statement about the budget configuration of Table 2, not about G-Eval with a frontier-grade judge; it is nonetheless the relevant statement for a practitioner who must run evaluation on their own hardware, and it illustrates why the cost axis changes conclusions that a correlation-only table would support.

*The frontier is sparse where deployment lives.* Between 3 ms and 430 ms — the latency range in which an online scorer for model selection or release gating actually has to operate — LGS is the frontier. That vacancy, rather than any absolute correlation height, is the claim of this paper.

#### 6.4. What the analysis shows and does not show

The Pareto reading supports the deployment thesis but does not by itself rank the metrics on quality: the y-axis is mean Spearman, which hides per-dimension variation (Section 7.4), and the x-axis is wall-clock runtime under our specific reproduction stack, not theoretical asymptotic complexity. Four caveats should be kept in mind. First, the LLM-judge cost depends linearly

Table 3: Cost–quality Pareto analysis of the fourteen metrics, ordered by per-sample runtime. Cost is wall-clock milliseconds per (source, candidate) pair under the controlled protocol of Section 5, summed over the four dimensions for dimensional scorers;  $\bar{\rho}$  is the mean summary-level Spearman across the four SUMMEVAL dimensions. A metric is dominated when another is no slower and no less correlated. Values are those of Figure 2 and Table 4.

| Metric            | Input         | ms / sample | $\bar{\rho}$ | Status                |
|-------------------|---------------|-------------|--------------|-----------------------|
| ROUGE-L           | reference     | 0.12        | .153         | Pareto-optimal        |
| SMART-String      | reference     | 0.52        | .084         | dominated by ROUGE-L  |
| BLEU              | reference     | 0.57        | .128         | dominated by ROUGE-L  |
| METEOR            | reference     | 0.98        | .162         | Pareto-optimal        |
| ChrF              | reference     | 2.77        | .184         | Pareto-optimal        |
| <b>LGS (ours)</b> | source        | 57.01       | .284         | <b>Pareto-optimal</b> |
| GPTScore          | reference     | 90.53       | .264         | dominated by LGS      |
| EMBEDSCORER       | reference     | 339.33      | .174         | dominated by LGS      |
| UniEval           | source + ref. | 430.31      | .470         | Pareto-optimal        |
| BARTScore         | reference     | 523.46      | .166         | dominated by LGS      |
| BERTScore         | reference     | 567.66      | .250         | dominated by LGS      |
| SMART-Model       | reference     | 890.02      | .202         | dominated by LGS      |
| MoverScore        | reference     | 1276.58     | .060         | dominated by LGS      |
| G-Eval            | source        | 13593.99    | .359         | dominated by UniEval  |

on the number of judge calls per pair, which we set at  $N=3$ ; a single-run G-Eval at  $N=1$  would shift left by a factor of three but would remain dominated by UniEval on both axes. Second, the embedding-metric costs depend on whether source-side embeddings are cached across candidates, which we do in LGS; without caching the per-pair cost would multiply by the candidate-to-source-sentence ratio, although in absolute terms it would still remain in

the embedding band rather than crossing into the LLM-judge band. Third, runtimes reflect a CPU-only single-thread deployment target; GPU serving would compress the differences between model-based metrics, though not the difference between one small embedder pass and  $N$  autoregressive decodes. Fourth, the mean over four dimensions is a summary statistic chosen for readability; a practitioner optimising a single dimension should read Table 4 directly. We discuss the deployment scenarios that put the frontier point to operational use in Section 8.1.

## 7. Results

Table 4 reports summary-level Spearman  $\rho$  correlations with human judgment for all fourteen metrics. Table 5 reports the paired-bootstrap deltas of LGS against each baseline on Spearman  $\rho$ . Table 6 reports system-level correlations (note that with  $n = 16$  systems the confidence intervals are wide; we treat this table as a secondary sanity check). The corresponding Kendall  $\tau$  tables are reported in Appendix Appendix A. The two coefficients induce almost the same ordering of the fourteen metrics: the rank correlation between the  $\rho$ -ordering and the  $\tau$ -ordering is .987–1.000 across the four dimensions, and no metric changes rank by more than two positions. The discussion below is therefore phrased in terms of  $\rho$  alone (Section 4.4).

### 7.1. Headline findings

*Ablation of the knowledge component.* The controlled test of the LGS design is the comparison against EMBEDSCORER, the whole-text cosine baseline on the same `nomic-embed-text` backbone (Section 2). LGS significantly outperforms EMBEDSCORER on coherence, consistency, and fluency ( $\Delta\rho = +$

Table 4: Summary-level correlations with human judgment on SummEval.  $\rho$  = Spearman. Brackets give the 95% cluster-bootstrap confidence interval over articles.

| Metric       | Coh                 | Con                 | Flu                | Rel               |
|--------------|---------------------|---------------------|--------------------|-------------------|
| BLEU         | .102 [.036, .165]   | .137 [.085, .185]   | .066 [.002, .123]  | .206 [.123, .277] |
| ROUGE-L      | .163 [.093, .231]   | .140 [.085, .195]   | .108 [.043, .167]  | .201 [.115, .278] |
| ChrF         | .158 [.092, .219]   | .174 [.125, .220]   | .110 [.050, .167]  | .293 [.219, .361] |
| METEOR       | .199 [.146, .250]   | .134 [.085, .179]   | .130 [.066, .193]  | .185 [.115, .251] |
| SMART-String | -.006 [-.070, .063] | .111 [.054, .166]   | .105 [.046, .159]  | .124 [.045, .194] |
| EmbedScorer  | .279 [.212, .339]   | .056 [-.001, .111]  | .041 [-.025, .107] | .320 [.245, .390] |
| BERTScore    | .377 [.317, .431]   | .109 [.050, .164]   | .142 [.077, .202]  | .371 [.301, .429] |
| MoverScore   | .152 [.085, .218]   | -.026 [-.084, .030] | .011 [-.054, .077] | .102 [.028, .179] |
| SMART-Model  | .328 [.267, .385]   | .093 [.035, .147]   | .067 [.000, .134]  | .321 [.248, .385] |
| BARTScore    | .167 [.102, .229]   | .117 [.055, .176]   | .106 [.037, .167]  | .273 [.192, .348] |
| GPTScore     | .358 [.308, .408]   | .210 [.142, .270]   | .181 [.109, .245]  | .306 [.247, .365] |
| UniEval      | .569 [.529, .606]   | .405 [.367, .443]   | .444 [.398, .485]  | .462 [.412, .509] |
| G-Eval       | .393 [.337, .444]   | .488 [.446, .531]   | .140 [.101, .177]  | .416 [.356, .473] |
| LGS          | .396 [.344, .449]   | .227 [.176, .275]   | .156 [.091, .220]  | .356 [.297, .407] |

.117/ + .171/ + .116, all  $p < .001$  by paired bootstrap with 5,000 resamples) and is in a statistical tie on relevance. Because the only difference between the two metrics is architectural — sentence-level grounding via  $\max_i$  on the source plus the lead-bias prior  $w(i)$ , versus a single whole-text cosine — the lift is directly attributable to those choices and cannot be explained by embedder strength. The prior itself is ablated separately in Section 7.2 by setting  $\lambda=0$ , which isolates the contribution of the injected structural knowledge from that of sentence-level grounding.

*Cross-family comparisons.* Against the broader baseline pool, LGS significantly outperforms SMART-Model on coherence, consistency, and fluency ( $\Delta\rho = +.068/ +.134/ +.089$ ,  $p \leq .005$ ), BERTScore on consistency ( $\Delta\rho = +.118$ ,  $p < .001$ ), and ChrF on coherence ( $\Delta\rho = +.237$ ,  $p < .001$ ). On the

Table 5: Paired bootstrap comparison of LGS against baselines on summary-level Spearman correlations. Each cell reports  $\Delta\rho$  (ours minus baseline) with 95% CI in brackets, the raw two-sided  $p$ -value, and  $q$ , the Benjamini-Hochberg adjusted  $p$ -value over all 24 (baseline, dimension) cells of this table. Bold marks improvements that survive the correction ( $q < 0.05$  and CI excludes 0).

| Baseline    | Coh                                                 | Con                                                 | Flu                                                 | Rel                                               |
|-------------|-----------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------|---------------------------------------------------|
| UniEval     | -.173<br>[-.236, -.113]<br>$p < .001$<br>$q < .001$ | -.177<br>[-.233, -.123]<br>$p < .001$<br>$q < .001$ | -.287<br>[-.355, -.226]<br>$p < .001$<br>$q < .001$ | -.107<br>[-.176, -.039]<br>$p=0.003$<br>$q=0.006$ |
| BERTScore   | +.019<br>[-.039, +.076]<br>$p=0.509$<br>$q=0.611$   | +.118<br>[+.059, +.181]<br>$p < .001$<br>$q < .001$ | +.015<br>[-.039, +.068]<br>$p=0.612$<br>$q=0.666$   | -.016<br>[-.081, +.052]<br>$p=0.634$<br>$q=0.666$ |
| G-Eval      | +.003<br>[-.076, +.080]<br>$p=0.938$<br>$q=0.938$   | -.262<br>[-.320, -.203]<br>$p < .001$<br>$q < .001$ | +.017<br>[-.055, +.086]<br>$p=0.638$<br>$q=0.666$   | -.061<br>[-.144, +.023]<br>$p=0.152$<br>$q=0.243$ |
| SMART-Model | +.068<br>[+.020, +.118]<br>$p=0.005$<br>$q=0.010$   | +.134<br>[+.084, +.182]<br>$p < .001$<br>$q < .001$ | +.089<br>[+.038, +.141]<br>$p < .001$<br>$q < .001$ | +.035<br>[-.022, +.095]<br>$p=0.248$<br>$q=0.331$ |
| ChrF        | +.237<br>[+.145, +.326]<br>$p < .001$<br>$q < .001$ | +.053<br>[-.013, +.121]<br>$p=0.123$<br>$q=0.211$   | +.046<br>[-.017, +.113]<br>$p=0.171$<br>$q=0.256$   | +.062<br>[-.031, +.153]<br>$p=0.188$<br>$q=0.265$ |
| EmbedScorer | +.117<br>[+.061, +.174]<br>$p < .001$<br>$q < .001$ | +.171<br>[+.122, +.222]<br>$p < .001$<br>$q < .001$ | +.116<br>[+.058, +.173]<br>$p < .001$<br>$q < .001$ | +.036<br>[-.026, +.101]<br>$p=0.283$<br>$q=0.357$ |

remaining dimensions of these comparisons LGS is in a statistical tie, and it exceeds BLEU, ROUGE-L, METEOR, SMART-String, MoverScore and BARTScore on the point estimate of every dimension. Against the top-tier



Table 6: System-level correlations with human judgment on SummEval.  $\rho$  = Spearman. Brackets give the 95% cluster-bootstrap confidence interval over articles.

| Metric       | Coh                 | Con                  | Flu                 | Rel                 |
|--------------|---------------------|----------------------|---------------------|---------------------|
| BLEU         | .276 [-.361, .720]  | .024 [-.589, .543]   | .415 [-.223, .818]  | .506 [-.134, .844]  |
| ROUGE-L      | .312 [-.324, .734]  | -.018 [-.607, .526]  | .365 [-.243, .771]  | .482 [-.139, .851]  |
| ChrF         | .526 [.022, .831]   | .794 [.505, .944]    | .832 [.555, .940]   | .803 [.498, .917]   |
| METEOR       | .094 [-.547, .582]  | -.318 [-.728, .229]  | .041 [-.538, .631]  | .179 [-.487, .641]  |
| SMART-String | -.029 [-.777, .511] | -.197 [-.656, .330]  | .171 [-.402, .679]  | .256 [-.468, .735]  |
| EmbedScorer  | .424 [-.097, .790]  | .406 [-.120, .796]   | .262 [-.296, .706]  | .262 [-.243, .745]  |
| BERTScore    | .768 [.406, .955]   | -.029 [-.590, .537]  | .324 [-.294, .818]  | .482 [-.071, .897]  |
| MoverScore   | .109 [-.532, .665]  | -.612 [-.874, -.050] | -.329 [-.836, .277] | -.115 [-.726, .543] |
| SMART-Model  | .700 [.242, .914]   | .594 [.033, .920]    | .538 [-.051, .905]  | .535 [-.056, .920]  |
| BARTScore    | .503 [-.006, .839]  | .812 [.517, .950]    | .785 [.484, .893]   | .776 [.471, .927]   |
| GPTScore     | .612 [.028, .955]   | .538 [-.015, .890]   | .379 [-.281, .858]  | .374 [-.259, .895]  |
| UniEval      | .791 [.407, .970]   | .444 [-.153, .863]   | .762 [.359, .928]   | .815 [.439, .949]   |
| G-Eval       | .526 [.019, .873]   | .902 [.646, .975]    | .259 [-.393, .812]  | .785 [.375, .970]   |
| LGS          | .529 [-.086, .958]  | .709 [.294, .871]    | .488 [-.179, .892]  | .453 [-.127, .943]  |

reference-free metrics, LGS loses: it is below UniEval on every dimension by paired bootstrap and below G-Eval specifically on consistency, while tying with G-Eval on coherence, fluency and relevance. We report this as the ceiling of an embedding-only metric of this size. The deployment claim of LGS is the Pareto position of Section 6.3, not the absolute correlation height.

*System-level correlations.* Table 6 reports system-level correlations for all metrics. With  $n=16$  systems the confidence intervals are wide and the per-cell rankings are unstable; we treat the table as a secondary sanity check, consistent with prior meta-evaluation practice on SUMMEVAL. The qualitative ordering of the metric families matches the summary-level table.

## 7.2. Ablating the knowledge prior: the exponent $\lambda$

Table 7 reports the development-split sweep of  $\lambda \in \{0, 0.25, 0.5, 1.0, 2.0\}$  and the test-split verification of the selected  $\lambda^* = 0.5$ . Setting  $\lambda=0$  removes the injected knowledge exactly, so the difference between the first row and the rest is the measured contribution of the prior. The mean Spearman  $\rho$  across the four dimensions is .233 at  $\lambda = 0$  (no prior) and .252 at  $\lambda^* = 0.5$ , a lift of +.019. On the held-out test split,  $\lambda^* = 0.5$  also outperforms the no-prior baseline on the mean (.319 versus .314). Beyond  $\lambda > 1$  the mean drops, indicating that the prior becomes too aggressive once the article tail is heavily suppressed.

The mean gap between  $\lambda = 0.25$  (.251) and  $\lambda^* = 0.5$  (.252) is .001, well inside the bootstrap noise. To characterise the surface around the optimum we re-ran a finer sweep  $\lambda \in \{0.30, 0.40, 0.50, 0.60, 0.75\}$  with paired bootstrap (5,000 resamples) on both splits. The development surface around the optimum is smooth and flat: per-dimension  $\rho$  values move by less than the bootstrap half-width across the entire  $[0.30, 0.75]$  window, and the test surface mirrors this pattern. On the combined nine-point grid the development argmax is in fact  $\lambda=0.4$  (.2528) rather than the  $\lambda^*=0.5$  (.2519) selected on the five-point grid of the pre-registered protocol; the two differ by .0009. We retain  $\lambda^*=0.5$  because the protocol of Section 4 fixes the grid before the sweep and the finer grid was added afterwards, but the discrepancy is itself evidence that the argmax is not a stable object — which the next section measures rather than asserts. We therefore read  $\lambda^* = 0.5$  as one representative of a flat plateau rather than an isolated peak.

The per-dimension picture on the test split is more nuanced than on the

development split. On development, all four SUMMEVAL dimensions improve at  $\lambda^* = 0.5$  over the no-prior baseline (Coh +.021, Con +.007, Flu +.011, Rel +.036). On test, only the mean and coherence carry that lift cleanly: Coh +.014, Con -.001, Flu +.001, Rel +.003. We therefore claim a meaningful held-out lift for the mean and for coherence; the apparent per-dimension lifts on consistency, fluency, and relevance are within bootstrap noise on the test split and we do not claim them as generalising effects of the lead-bias prior. We also note that the test-split argmax is  $\lambda=0.25$  (mean .334) rather than the development-selected  $\lambda^*=0.5$ ; under the protocol of Section 4 the test split does not participate in selection, and reporting the mismatch is part of the protocol’s value.

Beyond the optimum, the metric degrades gracefully but monotonically. At  $\lambda=1.0$  the development mean drops to .242 and at  $\lambda=2.0$  to .221 — below the no-prior baseline of .233. The mechanism is straightforward to see from the prior’s shape. At  $\lambda=1$ , the weight at the article tail  $w(n-1)$  is  $\exp(-1) \approx 0.37$ ; at  $\lambda=2$  it is  $\exp(-2) \approx 0.14$ . As  $\lambda$  grows, source sentences beyond the lead are increasingly suppressed before the  $\max_i$  operator, and a candidate sentence whose true semantic anchor is in the article body — a paraphrased detail, a quoted figure, a closing context — can no longer reach its anchor with enough weight to win the argmax. The argmax instead collapses onto mismatched lead sentences for these candidates, depressing cos, and  $\rho$  falls. This is the behaviour one should want from an injected prior: it helps when it gently up-weights the lead, and hurts when it functionally truncates the article. The mid-range  $\lambda \in [0.4, 0.6]$  is the operationally useful window, and the calibration sweep is what locates it.

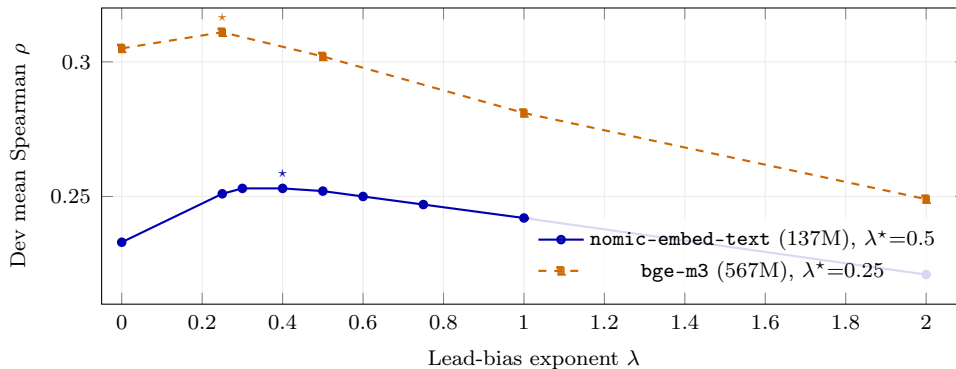


Figure 3: Development-split mean Spearman  $\rho$  across the four SUMMEVAL dimensions as a function of the lead-bias exponent  $\lambda$ , for the two sentence-embedder backbones whose sweep is reported in this paper. Both curves peak at  $\lambda > 0$  and decline monotonically past the peak, but the position of the peak *shifts* between backbones (0.5 for **nomic-embed-text**, 0.25 for **bge-m3**). Reusing the nomic-tuned  $\lambda=0.5$  with **bge-m3** would give  $\rho=0.302$ , below its no-prior baseline of 0.305. This is the empirical motivation for treating the calibration protocol of Section 4 as an obligatory part of the method.

Figure 4 decomposes the development-mean curve of Figure 3 (**nomic-embed-text**) into its four SUMMEVAL per-dimension components. The decomposition shows that the  $\lambda^*=0.5$  optimum is a compromise: the four dimensions do not share a single  $\lambda$  that is best for all of them. Coherence and relevance benefit strongly from the prior — coherence peaks broadly in the  $[0.25, 1.0]$  region, relevance continues to climb up to  $\lambda=2$  — while consistency and fluency are best at the *lowest* non-zero  $\lambda$  ( $\lambda=0.25$ ) and deteriorate sharply as  $\lambda$  grows. The mean optimum at  $\lambda=0.5$  is the value that best balances the two pairs; practitioners with a single dimension of interest may prefer a different point on the curve.

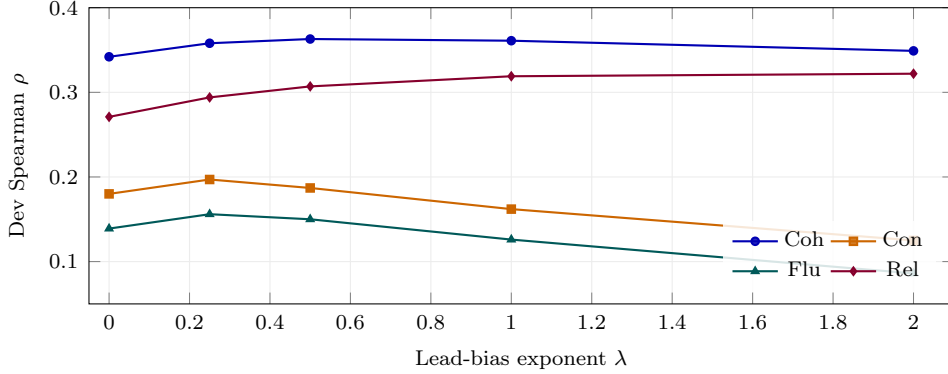


Figure 4: Per-dimension development-split Spearman  $\rho$  versus the lead-bias exponent  $\lambda$  with `nomic-embed-text`. Coherence and relevance benefit from the prior across the swept range ( $\lambda_{\text{Coh}}^*=0.5$ ,  $\lambda_{\text{Rel}}^*=2.0$ ); consistency and fluency peak at the smallest non-zero value ( $\lambda_{\text{Con}}^*, \lambda_{\text{Flu}}^*=0.25$ ) and degrade as  $\lambda$  grows. The canonical  $\lambda^*=0.5$  is the optimum on the mean-of-four criterion (Section 4), not on any single dimension.

### 7.3. How identifiable is $\lambda^*$ ?

Every number reported so far treats  $\lambda^*$  as if the development split had determined it. It did not determine it; it selected it, and a different draw of 50 articles would have selected something else. Because the entire calibration protocol of Section 4 rests on that selection, we measure how reproducible it is instead of assuming it.

The procedure is the cluster bootstrap of Section 4.5 applied one level up. On each of 5,000 resamples of the development articles we recompute the mean Spearman  $\rho$  at *every* grid value and record which value wins, yielding a distribution over the grid rather than a point. Table 8 reports it. Because the per-sample scores for each  $\lambda$  are already stored, the measurement requires no additional embedder passes.

*Switching the prior on is reproducible.* The no-prior setting  $\lambda=0$  is selected in 0.7% of development resamples and 0.3% of test resamples. Whatever else is uncertain, the decision to impose *some* positional prior is not: it survives resampling in more than 99% of draws on both splits. This is the claim the ablation supports, and it is weaker than — and logically prior to — any claim about a particular exponent.

*The exponent itself is not identifiable.* The selection mass does not concentrate. On the development split with the canonical backbone,  $\lambda=0.3$  wins 27.6% of resamples,  $\lambda=0.4$  wins 20.4%, the reported  $\lambda^*=0.5$  wins 17.8%, and  $\lambda=0.25$  wins 17.5%: four grid values within ten points of each other, none commanding even a third of the mass. A 50-article development split does not contain enough information to separate them. The practical reading is that the protocol returns the interval  $[0.25, 0.6]$ , inside which no value is distinguishable from any other, and that quoting a three-significant-figure optimum from such a surface overstates what was measured.

*Two controls, and what they do to the backbone claim.* The lower panel of Table 8 compares whole selection outcomes rather than single  $\rho$  values, on the grid common to every configuration. The control that matters is the third row of the reasoning: a configuration compared *against itself* on two independent draws of the same 50 articles reselects the same exponent only 43.5% of the time, and when it disagrees the direction is a coin flip (28.6% versus 27.9%). That is the noise floor. Against it, the cross-backbone agreement of 37.8% is barely lower — so the observation that two backbones pick different exponents is, on its own, uninformative.

What does separate from the noise floor is the *direction*. When `nomic-embed-text` and `bge-m3` are scored on the same resample and disagree, the former selects the larger exponent in 61.9% of resamples against 0.4% for the reverse, a ratio no symmetric noise process produces. We therefore claim a directional backbone effect — `bge-m3` prefers a weaker prior — and explicitly do not claim a quantified shift from 0.5 to 0.25.

One confound must be stated because it limits even that claim. The same asymmetry appears between the development and test splits under an *unchanged* backbone (52.8% versus 0.9%). Changing which 50 articles are used therefore induces a directional shift of comparable strength to changing the representation space. Our design cannot attribute the cross-backbone asymmetry to the backbone alone, and a clean attribution would need the sweep repeated over many disjoint article samples per backbone — which, on a 100-article benchmark, is not available. Section 8.6 lists the multi-corpus replication that would supply it.

#### 7.4. Backbone ablation

Table 9 shows the canonical  $\lambda=0.5$  metric with four sentence-embedder backbones spanning a  $\sim 24\times$  parameter range: `all-minilm` (23M), `nomic-embed-text` (137M), `mxbai-embed-large` (335M), and `bge-m3` (567M). The mean Spearman across the four SUMMEVAL dimensions falls in  $[\text{.284}, \text{.312}]$  for all four backbones, a span of .028 across the full parameter range. We read this as evidence that the metric *design* — the mean-of-max grounding aggregator with the lead-bias prior — is qualitatively robust to backbone choice: any reasonable sentence embedder produces a working metric.

Two properties of this table need to be stated explicitly, because both

cut against the headline numbers rather than for them. First, **nomic-embed-text** — the canonical backbone, fixed before the ablation was run — has the *lowest* mean of the four (.284, against .296 for **mxbai-embed-large**, .298 for **all-minilm** and .312 for **bge-m3**). The headline correlations of Tables 4 and 5, and the Pareto position of Section 6.3, are therefore reported at the conservative end of the observed band; a backbone swap to **bge-m3** would raise the mean  $\rho$  to .312 without changing the design, at roughly 5 $\times$  the embedding cost. We keep the pre-registered canonical backbone rather than the best-performing one, because selecting the backbone on the same data used for the headline table would be a researcher degree of freedom we would rather not exercise.

Second, the per-dimension profile is not stable across backbones. **nomic-embed-text** is by far the strongest on coherence (.396) and relevance (.356) and the weakest on consistency (.227) and fluency (.156); the other three backbones invert this pattern. *Per-dimension* comparisons against baselines — for instance, “LGS exceeds BERTScore on consistency” — are therefore valid for the specific backbone used and should be read as backbone-conditional, not as universal properties of LGS. The mean-Spearman ordering of metric families is stable across backbones; the per-dimension ordering is not.

### 7.5. Cross-backbone calibration of the prior

The backbone ablation also bears on *where* the prior should be set, not only on how well the metric scores. Figure 3 shows the development-split mean Spearman  $\rho$  as a function of  $\lambda$  for two embedders: **nomic-embed-text** (the canonical backbone) and **bge-m3** (the largest in our roster, chosen as



the strongest contrast point). Both curves share their qualitative shape: each peaks at  $\lambda > 0$  — that is, the injected structural knowledge helps regardless of the backbone — and declines monotonically past the peak. But the position of the peak *shifts*:  $\lambda^* = 0.5$  for `nomic-embed-text` (development mean .252, a lift of +.019 over the no-prior baseline) and  $\lambda^* = 0.25$  for `bge-m3` (development mean .311, a lift of only +.006).

The practical consequence is sharp: applying the nomic-tuned  $\lambda = 0.5$  to `bge-m3` yields a development mean of .302, which is *below* its no-prior baseline of .305. A prior that helps when its exponent is calibrated against the backbone in use can therefore cost more than it returns when an exponent is carried over from elsewhere.

Two qualifications are essential, and Section 7.3 supplies both. First, the gap between .302 and .305 is a fraction of the bootstrap half-width, so this particular pair of numbers is an illustration rather than a demonstration. What carries statistical weight is the directional result: scored on the same resample, `nomic-embed-text` selects the larger exponent 61.9% of the time against 0.4% for the reverse. Second, that asymmetry is not uniquely attributable to the backbone, because a change of article sample under a fixed backbone produces one of comparable strength. The defensible claim is therefore that the exponent should be re-derived in situ — not that the backbone is the specific variable that invalidates a borrowed value. That weaker claim is sufficient for the protocol, since the sweep is what transfers and it costs no additional embedder passes.

We hypothesise that the size of the lift depends on the density of the backbone’s representation. `bge-m3` already produces a stronger position-

agnostic recall (.305 development mean at  $\lambda=0$ , versus .233 for `nomic-embed-text`), leaving less room for a positional prior to add value. We flag this as a hypothesis rather than a result: with two backbones swept and an exponent that Section 7.3 shows to be poorly identified, the evidence does not discriminate between it and simpler accounts. The general lesson we do draw is weaker and, we think, more durable: an a priori structural prior and a learned representation are not independent components, and how much the explicit prior can still contribute is not knowable without measurement on the deployment’s own data.

Table 7: Ablation of LGS on SummEval (summary-level Spearman  $\rho$ ). The metric is  $\text{score} = \text{mean}_j \max_i w(i) \cdot \cos(c_j, s_i)$  with exponential lead-bias prior  $w(i) = \exp(-\lambda \cdot i/n)$  on source-sentence position. The exponent  $\lambda$  is selected on a held-out development split (50 articles) and re-evaluated on the test split (50 articles); the dev-selected canonical row is marked  $\star$ .

| Variant                                      | Coh  | Con  | Flu  | Rel  |
|----------------------------------------------|------|------|------|------|
| Recall baseline ( $\lambda = 0$ ) (dev)      | .342 | .180 | .139 | .271 |
| Lead-bias ( $\lambda = 0.25$ ) (dev)         | .358 | .197 | .156 | .294 |
| Lead-bias ( $\lambda = 0.3$ ) (dev)          | .361 | .196 | .156 | .298 |
| Lead-bias ( $\lambda = 0.4$ ) (dev)          | .363 | .191 | .153 | .303 |
| Lead-bias ( $\lambda = 0.5$ ) (dev)          | .363 | .187 | .150 | .307 |
| Lead-bias ( $\lambda = 0.6$ ) (dev)          | .363 | .183 | .145 | .311 |
| Lead-bias ( $\lambda = 0.75$ ) (dev)         | .362 | .175 | .138 | .315 |
| Lead-bias ( $\lambda = 1$ ) (dev)            | .361 | .162 | .126 | .319 |
| Lead-bias ( $\lambda = 2$ ) (dev)            | .349 | .125 | .086 | .322 |
| Recall baseline ( $\lambda = 0$ ) (test)     | .415 | .272 | .161 | .410 |
| Lead-bias ( $\lambda = 0.25$ ) (test)        | .433 | .295 | .180 | .427 |
| Lead-bias ( $\lambda = 0.3$ ) (test)         | .434 | .293 | .178 | .425 |
| Lead-bias ( $\lambda = 0.4$ ) (test)         | .432 | .282 | .171 | .419 |
| Lead-bias ( $\lambda = 0.5$ ) (test) $\star$ | .429 | .271 | .162 | .413 |
| Lead-bias ( $\lambda = 0.6$ ) (test)         | .425 | .261 | .152 | .404 |
| Lead-bias ( $\lambda = 0.75$ ) (test)        | .418 | .248 | .139 | .390 |
| Lead-bias ( $\lambda = 1$ ) (test)           | .402 | .227 | .121 | .369 |
| Lead-bias ( $\lambda = 2$ ) (test)           | .375 | .158 | .079 | .336 |

Table 8: Selection uncertainty of the lead-bias exponent. For each (split, backbone) the whole selection procedure — compute mean Spearman  $\rho$  across the four SUMMEVAL dimensions at every grid value, take the argmax — is re-run on 5000 cluster-bootstrap resamples over articles.  $P(\arg \max)$  is the fraction of resamples in which that grid value is selected; a selection procedure that identifies a well-separated optimum concentrates this mass on one row.  $\lambda^*$  marks the argmax of the point estimate, i.e. the value the ablation table reports.

| Split / backbone                                   | $\lambda$    | Mean $\rho$                    | 95% CI                         | $P(\arg \max)$                 |
|----------------------------------------------------|--------------|--------------------------------|--------------------------------|--------------------------------|
| dev / <b>bge-m3</b>                                | 0            | .305                           | [.248, .361]                   | 13.8%                          |
|                                                    | 0.25 $\star$ | .311                           | [.254, .366]                   | 84.2%                          |
|                                                    | 0.5          | .302                           | [.247, .355]                   | 2.0%                           |
|                                                    | 1            | .281                           | [.231, .330]                   | 0.0%                           |
|                                                    | 2            | .249                           | [.204, .295]                   | 0.0%                           |
| dev / <b>nomic-embed-text</b>                      | 0            | .233                           | [.168, .299]                   | 0.7%                           |
|                                                    | 0.25         | .251                           | [.188, .315]                   | 17.5%                          |
|                                                    | 0.3          | .253                           | [.189, .315]                   | 27.6%                          |
|                                                    | 0.4 $\star$  | .253                           | [.189, .315]                   | 20.4%                          |
|                                                    | 0.5          | .252                           | [.189, .314]                   | 17.8%                          |
|                                                    | 0.6          | .250                           | [.188, .312]                   | 9.9%                           |
|                                                    | 0.75         | .247                           | [.186, .308]                   | 2.6%                           |
|                                                    | 1            | .242                           | [.182, .301]                   | 2.9%                           |
|                                                    | 2            | .221                           | [.165, .276]                   | 0.5%                           |
| test / <b>nomic-embed-text</b>                     | 0            | .314                           | [.273, .352]                   | 0.3%                           |
|                                                    | 0.25 $\star$ | .334                           | [.291, .371]                   | 85.0%                          |
|                                                    | 0.3          | .332                           | [.289, .370]                   | 14.6%                          |
|                                                    | 0.4          | .326                           | [.282, .365]                   | 0.1%                           |
|                                                    | 0.5          | .319                           | [.274, .359]                   | 0.0%                           |
|                                                    | 0.6          | .311                           | [.265, .352]                   | 0.0%                           |
|                                                    | 0.75         | .299                           | [.252, .340]                   | 0.0%                           |
|                                                    | 1            | .280                           | [.228, .325]                   | 0.0%                           |
|                                                    | 2            | .237                           | [.182, .288]                   | 0.0%                           |
| Paired comparison                                  |              | $P(\lambda_A^* = \lambda_B^*)$ | $P(\lambda_A^* < \lambda_B^*)$ | $P(\lambda_A^* > \lambda_B^*)$ |
| cross-backbone (shared draw)                       |              | 37.8%                          | 0.4%                           | 61.9%                          |
| same backbone, two independent draws (noise floor) |              | 43.5%                          | 28.6%                          | 27.9%                          |
| dev vs test, same backbone                         |              | 46.3%                          | 0.9%                           | 52.8%                          |

Table 9: Embedder ablation: canonical LGS ( $\lambda = 0.5$ ) on the full SummEval set with four sentence-embedder backbones (Ollama). Reports summary-level Spearman  $\rho$  per dimension and the mean across the four. Canonical embedder (used in the headline tables) marked  $\star$ .

| Embedder                        | Coh  | Con  | Flu  | Rel  | Mean |
|---------------------------------|------|------|------|------|------|
| nomic-embed-text (137M) $\star$ | .396 | .227 | .156 | .356 | .284 |
| mxbai-embed-large (335M)        | .304 | .351 | .263 | .266 | .296 |
| all-minilm (23M)                | .291 | .367 | .276 | .259 | .298 |
| bge-m3 (567M)                   | .335 | .360 | .251 | .299 | .312 |

## 8. Discussion

### 8.1. Decision-support deployment scenarios

LGS is designed for three decision procedures that reference-based and LLM-judge metrics serve poorly. The first is *online model selection*: a summarisation system serves many candidates per source article (multiple decoding configurations, multiple prompt variants, multiple checkpoints under A/B test), and the operator needs a low-latency, deterministic score to pick the best candidate per request. Reference-based metrics are unavailable because there is no human reference for production inputs; G-Eval’s per-candidate judge call is too expensive and its sampled decoding too non-deterministic to make ranking reproducible across replays. LGS is reference-free, deterministic up to the embedder’s numerical precision, and adds approximately one CPU sentence-embedder forward pass per source and per candidate sentence to the request budget — with the source side amortised across candidates (Section 3.2).

The second is *automated release gating*. A continuous-integration pipeline replays a fixed corpus of source articles through both the previous and candidate model versions and blocks a merge on a significant regression. Such pipelines run at high frequency (per commit, per nightly build) over many thousands of articles; the cost of a judge or fine-tuned-encoder pass per (article, candidate) pair under this load is prohibitive. The reference-free, embedding-only profile of LGS makes the wall-clock overhead negligible, and its single hyperparameter is fixed at deployment time, so the score is a stable signal across runs. Determinism matters here beyond cost: a gate that fires non-deterministically is not a gate.

The third is *reward modelling for reinforcement-learning fine-tuning of summarisation systems*. Such training generates many candidates per article per step, requiring a fast and stable reward signal. LGS produces a deterministic scalar per candidate from a single embedder pass, which is compatible with the iteration speed demanded by these training loops, and the source-side embedding cache is reused across the whole rollout for a given article. We do not claim that LGS is the ideal reward signal — specialised reward models tuned to the target dimension would likely dominate — but it is a viable default when no such specialised model exists, particularly on news-domain corpora where the injected prior matches the underlying distribution.

In all three cases the metric is consumed by an automated procedure that must justify its decision to a human afterwards. The argmax anchors of Algorithm 1 serve that purpose: a rejected candidate can be reported together with the specific summary sentences that failed to find support in the source and the source sentences that came closest, which neither a whole-text cosine nor a judge’s numeric verdict provides without extra work.

## 8.2. What LGS does and does not claim

We make four claims and explicitly disclaim the stronger versions of each. (i) LGS is Pareto-optimal in the cost–quality plane of our controlled benchmark and strictly dominates six widely used metrics; we do *not* claim the highest correlation — UniEval is higher on every dimension and we report that directly. (ii) The injected lead-bias prior produces a held-out lift on the *mean* Spearman  $\rho$  across the four SUMMEVAL dimensions and on *coherence* specifically; we do *not* claim a held-out per-dimension lift on consistency, fluency, or relevance, where the development-side improvements

(+.007, +.011, +.036 respectively) attenuate to within bootstrap noise on test (−.001, +.001, +.003). (iii) Imposing some positional prior is supported:  $\lambda=0$  is selected in under 1% of bootstrap resamples on either split. We do *not* claim that  $\lambda^* = 0.5$  is the optimum, universal or otherwise — it wins 17.8% of development resamples, three other grid values win comparable shares, and an unchanged configuration reselects it only 43.5% of the time across two draws. What the protocol returns is the interval  $[0.25, 0.6]$  (Section 7.3). We further claim a *directional* backbone effect — **bge-m3** prefers a weaker prior — but not a quantified shift, and we note that a change of article sample under a fixed backbone produces a directional shift of comparable strength, so the backbone is not isolated as the cause (Section 7.5). (iv) The cost figures characterise a single-machine, CPU-only, single-thread deployment target; we do *not* claim that the same ratios hold under GPU serving or batched inference, although the structural difference between one small embedder pass and  $N$  autoregressive decodes is not a property of the hardware.

### 8.3. Practitioner recommendations

For readers considering LGS as a deployed metric, the empirical findings above translate into four operational recommendations. First, *run the calibration sweep*. The optimal  $\lambda$  depends on the backbone (Section 7.5), and reusing a published  $\lambda$  value from a different backbone can degrade the metric below its no-prior baseline. The sweep is cheap — five  $\lambda$  values on a 50-article development split take approximately five minutes on the canonical **nomic-embed-text** backbone, and require no additional embedder passes beyond the first (Section 3.2) — and is the operational price of an explicit



structural prior. Second, *decide whether the mean or a single dimension is the deployment target*. The canonical  $\lambda^*=0.5$  optimises the mean across the four SUMMEVAL dimensions; if the deployment cares specifically about coherence or relevance the optimum shifts toward larger  $\lambda$ , and if it cares about consistency or fluency the optimum shifts toward the smallest non-zero  $\lambda$  in the grid (Figure 4). Third, *pick the backbone on the deployment’s cost budget, not on ours*. The backbone ablation (Section 7.4) shows the mean  $\rho$  varies by only .028 across a  $24\times$  parameter range, so a deployment that can afford **bge-m3** gains a slightly higher mean, and one that cannot loses very little by using a 23M-parameter embedder — but the per-dimension profile changes substantially, so the choice should be made against the dimension that matters. Fourth, *re-tune  $\lambda$  when migrating backbones*. A backbone change is not a drop-in replacement; it is a new deployment from the metric’s perspective.

#### 8.4. Limitations

*Domain.* SUMMEVAL contains only CNN/DailyMail news articles. Lead bias is a well-documented structural property of this corpus, and LGS exploits it explicitly through the exponential prior. Domains without a strong leading-sentence structure — scientific abstracting, narrative or dialogue summarisation, long-form question answering — may require a different prior or none at all. We do not claim that the exponential lead-bias formulation transfers; we claim that the *protocol* of selecting and calibrating a structural source-side prior on a held-out split transfers, and that the prior shape itself is a domain-specific design choice.

*G-Eval baseline configuration.* Our G-Eval baseline uses  $N = 3$  independent runs at temperature 0.7 with **qwen2.5:7b-instruct-q4\_K\_M** and

reports the across-run mean as the canonical score. The original paper averages 20 GPT-4 samples; our setup is therefore a budget reproduction. The across-run standard deviation of summary-level Spearman is between .003 and .009 across the four dimensions, well below the half-width of the paired-bootstrap confidence intervals, so judge stochasticity is not a meaningful confound for the LGS-vs-G-Eval comparisons. We document this explicitly because single-run G-Eval baselines in the literature do not.

*Prior shape.* The lead-bias prior is exponential. Linear, reciprocal, and hard-cutoff alternatives have not been tested; we leave them as future work (Section 8.6).

*Backbone.* The headline numbers use `nomic-embed-text` with  $\lambda^*$  calibrated against it. The ablation in Section 7.4 shows that mean Spearman is qualitatively stable across a  $\sim 24\times$  parameter range, but per-dimension profiles shift; numbers quoted at the per-dimension granularity should be read as backbone-conditional.

### 8.5. Threats to validity

We flag four threats to the validity of the conclusions drawn in this paper that a careful reader should weigh.

*Single-corpus evidence.* All correlation numbers are computed on SUMMEVAL, a single benchmark derived from a single news corpus (CNN/DailyMail). The *methodological* contributions — the article-level held-out split, the cluster-bootstrap discipline, the cost-controlled protocol, and the demonstration that the optimal exponent shifts across backbones — are corpus-independent in principle, but the *numerical* headline correlations transfer to other corpora only to the degree that those corpora share

the lead-bias structural property of SUMMEVAL articles. We consider a multi-corpus replication the single most valuable extension of this work (Section 8.6).

*Reproductions versus original-paper numbers.* As discussed in Section 5 and Table 2, our baseline numbers are not the original-paper numbers. They are reproductions under budget-feasible configurations on a single laptop-scale deployment target, which is what makes the cost axis meaningful. Cross-paper number comparisons should therefore be made against the original references, not against our reproductions; our reproductions are internally self-consistent for the cross-baseline comparisons we report, and the Pareto structure of Section 6.3 is a statement about this internally consistent pool.

*Stochasticity of the LLM judge.* Our G-Eval configuration uses  $N=3$  samples at temperature 0.7. The across-run standard deviation of summary-level Spearman is between .003 and .009, which we have argued is small enough not to be a meaningful confound for the LGS-vs-G-Eval comparisons (Section 8.4). A full  $N=20$  reproduction at temperature 1 with a frontier-grade judge would shrink this further and would raise G-Eval’s correlation, but at a proportionally higher cost, so it would not move G-Eval onto the Pareto frontier unless the correlation gain exceeded UniEval’s.

*Backbone calibration of the score scale.* The cosine similarity values used inside LGS are not calibrated probabilities; they are raw inner products in a learned representation space. The metric is therefore meaningful as a *ranker* of summaries (which is what Spearman measures) but should not be interpreted as a calibrated quality score on an absolute scale.

### 8.6. Future work

Four directions extend the current contribution most directly. First, transfer to non-news corpora, which is where the injected knowledge is most at risk: scientific abstracting (ARXIV, PUBMED, see [41]) and dialogue summarisation (SAMSUM [42], DIALOGSUM) are natural next benchmarks, and the prior shape is the variable we expect to re-tune — or to remove, if the corpus has no positional salience structure. Second, alternative shapes for the source-position prior: linear  $w(i) = \max(0, 1 - \lambda i/n)$ , reciprocal  $w(i) = 1/(1 + \lambda i/n)$ , and a hard cutoff at the first  $k$  sentences. The calibration protocol of Section 4 applies unchanged. Third, richer sources of structural knowledge than position: salience weighting derived from discourse parsing, section headings in structured documents, or entity-centrality signals over the source — all of which fit the same multiplicative source-side slot as  $w(i)$  and can be ablated the same way. Fourth, alternative aggregation operators on the candidate side: low-percentile recall (a hallucination-sensitive variant) and position-weighted top- $k$  recall, evaluated against the diffusion pattern documented in Appendix Appendix B.

## 9. Conclusion

We presented LGS, a reference-free, embedding-only operator for summary quality assessment, defined as  $\text{LGS}(D, C) = \text{mean}_j \max_i w(i) \cos(c_j, s_i)$  with an exponential lead-bias prior  $w(i) = \exp(-\lambda i/n)$  over source-sentence position. The prior is the point of the design: it injects one explicit, auditable piece of discourse-structural domain knowledge into a subsymbolic similarity space, at zero inference cost and with a single interpretable parameter that

can be set to zero to remove it exactly.

Measured against thirteen baselines on SUMMEVAL under one controlled hardware budget, LGS is one of only five Pareto-optimal metrics in the cost–quality plane, and it strictly dominates six widely used metrics — among them BERTScore, whose mean summary-level Spearman  $\rho$  it exceeds while running roughly ten times faster and without access to the human reference summary. In the same analysis the LLM-judge baseline is itself dominated by the fine-tuned-encoder baseline — a statement about the budget configurations of Table 2, not about G-Eval with a frontier-grade judge, and one we make only for the reproduction pool defined in Section 5. Even so restricted, it is a conclusion that no correlation-only table can reach. We are explicit that UniEval attains a higher correlation than LGS on every dimension; the claim of this paper is a position in the cost–quality plane, not a correlation record.

The second contribution is methodological and, we believe, transfers beyond this metric. Submitting the hyperparameter *selection procedure* to the same bootstrap that validates the correlations separates two claims that a point estimate conflates. That the prior should be switched on is reproducible: the no-prior setting is selected in under 1% of resamples. Which exponent to switch it to is not: the reported value wins under a fifth of resamples, and an unchanged configuration reselects it less than half the time across two draws of the same articles. A knowledge-informed metric of this kind should therefore be published as a *design plus an obligatory calibration protocol* that returns an interval, not as a fixed numerical recipe — and the width of that interval belongs in the paper, because a practitioner who

copies a published exponent is otherwise copying one draw’s answer to an underdetermined question. The calibration is cheap — five  $\lambda$  values on 50 articles, approximately five minutes on a laptop CPU thread, with no additional embedder passes — and the cost of skipping it is silent miscalibration. The same discipline applies to alternative priors and alternative aggregation operators; Appendix Appendix B records the components we tested and rejected, together with the structural pattern extracted from those negative results.

## Reproducibility

The full pipeline is open-source and deterministic up to the sentence embedder’s numerical precision. The repository (<https://github.com/mikolajsemeniuk/llmbench>) contains a Makefile reproducing every table and figure in this paper end-to-end: `make benchmark` runs all 14 metrics on SUMMEVAL, `make benchmark-ablation` runs the  $\lambda$  sweep on the development and test splits, `make benchmark-embedder-ablation` runs the canonical metric with the four backbones, and `make paper` regenerates every `.tex` table from the corresponding JSON outputs. The selection-uncertainty analysis of Section 7.3 is `make paper-lamdaci`, and the multiple-comparison correction of Section 4.5 is applied inside `cmd/compare`; both read the stored per-sample scores and need neither Ollama nor the model server, so a reviewer can reproduce every statistical claim in this paper without running a single metric. Bootstrap seeds are fixed in `pkg/eval` (seed 42 for cluster bootstrap, seed-derived for paired comparisons). The sentence embedders are pinned by Ollama tag in the

Makefile (`nomic-embed-text`, `mxbai-embed-large`, `bge-m3`, `all-minilm`); the G-Eval judge is `qwen2.5:7b-instruct-q4_K_M` under the same Ollama runtime. SUMMEVAL itself is publicly available under the original authors’ release [9]; the JSONL split used for the article-level development/test partition is the SummEval canonical release order, deterministic without a random seed.

### **CRedit authorship contribution statement**

**Mikolaj Semeniuk:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. **Agnieszka Jastrzebska:** Conceptualization, Methodology, Validation, Supervision, Project administration, Writing – review & editing.

### **Declaration of competing interest**

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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### **Data availability**

All data and code required to reproduce every result in this paper are publicly available. The source code, the reproduction `Makefile`, and the per-

sample score files for all fourteen metrics are released at <https://github.com/mikolajsemeniuk/llmbench>. The SUMMEVAL benchmark is publicly distributed by its original authors [9].

### **Declaration of generative AI and AI-assisted technologies in the writing process**

During the preparation of this work the authors used a generative AI assistant in order to improve the language and readability of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

### **Appendix A. Kendall $\tau$ Correlations**

Section 4.4 states that Spearman  $\rho$  is the primary correlation coefficient of this study and Kendall  $\tau$  a secondary check. Tables A.10 and A.11 report the  $\tau$  counterparts of Tables 4 and 6, with the same article-level cluster bootstrap. The two coefficients induce almost the same ordering of the fourteen metrics — the rank correlation between the  $\rho$ -ordering and the  $\tau$ -ordering is .996 on coherence, 1.000 on consistency, .987 on fluency and .996 on relevance, and no metric changes rank by more than two positions on any dimension. The conclusions of Section 7 are therefore invariant to the choice between the two rank coefficients.



Table A.10: Summary-level correlations with human judgment on SummEval.  $\tau =$  Kendall. Brackets give the 95% cluster-bootstrap confidence interval over articles.

| Metric       | Coh                 | Con                 | Flu                | Rel               |
|--------------|---------------------|---------------------|--------------------|-------------------|
| BLEU         | .072 [.024, .115]   | .107 [.065, .147]   | .051 [.005, .097]  | .148 [.094, .203] |
| ROUGE-L      | .115 [.067, .162]   | .110 [.062, .154]   | .083 [.036, .134]  | .144 [.084, .202] |
| ChrF         | .112 [.065, .160]   | .137 [.101, .174]   | .086 [.043, .132]  | .210 [.158, .264] |
| METEOR       | .139 [.100, .176]   | .105 [.064, .143]   | .101 [.051, .151]  | .131 [.078, .179] |
| SMART-String | -.004 [-.051, .043] | .087 [.043, .130]   | .081 [.037, .123]  | .088 [.034, .143] |
| EmbedScorer  | .197 [.153, .238]   | .044 [-.004, .088]  | .031 [-.018, .081] | .229 [.179, .279] |
| BERTScore    | .269 [.225, .307]   | .085 [.038, .128]   | .110 [.063, .156]  | .270 [.224, .318] |
| MoverScore   | .106 [.059, .150]   | -.020 [-.066, .026] | .009 [-.045, .057] | .072 [.016, .125] |
| SMART-Model  | .233 [.191, .273]   | .074 [.029, .116]   | .052 [.004, .103]  | .229 [.182, .277] |
| BARTScore    | .117 [.075, .161]   | .092 [.046, .137]   | .083 [.033, .132]  | .195 [.141, .252] |
| GPTScore     | .256 [.218, .293]   | .166 [.119, .216]   | .142 [.096, .194]  | .221 [.175, .264] |
| UniEval      | .408 [.378, .437]   | .323 [.293, .353]   | .352 [.317, .385]  | .334 [.293, .370] |
| G-Eval       | .318 [.276, .361]   | .448 [.410, .484]   | .131 [.094, .165]  | .345 [.293, .398] |
| LGS          | .285 [.243, .326]   | .179 [.137, .219]   | .120 [.076, .169]  | .257 [.216, .298] |

Table A.11: System-level correlations with human judgment on SummEval.  $\tau =$  Kendall. Brackets give the 95% cluster-bootstrap confidence interval over articles.

| Metric       | Coh                 | Con                  | Flu                 | Rel                 |
|--------------|---------------------|----------------------|---------------------|---------------------|
| BLEU         | .183 [-.276, .607]  | .017 [-.407, .459]   | .283 [-.228, .691]  | .350 [-.146, .709]  |
| ROUGE-L      | .217 [-.243, .636]  | -.017 [-.477, .424]  | .250 [-.183, .648]  | .317 [-.143, .712]  |
| ChrF         | .350 [-.045, .673]  | .617 [.351, .852]    | .617 [.357, .835]   | .617 [.375, .804]   |
| METEOR       | .000 [-.472, .404]  | -.200 [-.568, .264]  | .033 [-.439, .523]  | .067 [-.439, .486]  |
| SMART-String | -.150 [-.655, .297] | -.117 [-.469, .268]  | .117 [-.299, .535]  | .150 [-.386, .593]  |
| EmbedScorer  | .350 [.009, .646]   | .283 [-.091, .630]   | .183 [-.232, .540]  | .217 [-.171, .596]  |
| BERTScore    | .617 [.254, .887]   | -.017 [-.473, .460]  | .250 [-.232, .702]  | .383 [-.086, .796]  |
| MoverScore   | .083 [-.416, .541]  | -.450 [-.764, -.045] | -.283 [-.707, .174] | -.150 [-.655, .345] |
| SMART-Model  | .533 [.164, .802]   | .467 [.055, .790]    | .400 [-.089, .784]  | .467 [-.027, .874]  |
| BARTScore    | .367 [-.064, .694]  | .633 [.382, .870]    | .567 [.315, .784]   | .600 [.327, .835]   |
| GPTScore     | .500 [.053, .842]   | .400 [-.018, .762]   | .300 [-.236, .770]  | .367 [-.141, .852]  |
| UniEval      | .650 [.298, .911]   | .367 [-.057, .750]   | .583 [.252, .836]   | .633 [.321, .844]   |
| G-Eval       | .400 [-.018, .752]  | .745 [.491, .909]    | .226 [-.271, .661]  | .650 [.256, .927]   |
| LGS          | .467 [.009, .892]   | .467 [.135, .735]    | .400 [-.088, .807]  | .433 [-.035, .861]  |

## Appendix B. Extension Log: Components Tested and Rejected

The canonical LGS reported in this paper has a single tunable hyperparameter,  $\lambda$ . This minimal design is the result of a sequence of ablations conducted during calibration; each rejected component below was tested under the same held-out protocol of Section 4 (sweep on the 50-article development split, mean Spearman  $\rho$  across the four SUMMEVAL dimensions as the selection criterion, verification on the 50-article test split). We document them here so that readers can see what was tried, why it was rejected, and what general lesson the rejection encoded.

Table B.12 summarises the development-split mean Spearman  $\rho$  at the optimum of each rejected component’s grid, against the no-prior recall baseline ( $\rho=.233$ ) and the accepted lead-bias prior ( $\rho=.252$  at  $\lambda^*=0.5$ ).

*Pattern across rejections.* Three of the four rejected components — coverage, top- $k$  recall, and within-summary diversity — share a structural feature: each multiplies the recall term by a value in  $[0, 1]$  that *diffuses* the matching signal across multiple candidate–source pairings. The empirical pattern is consistent: each diffusion-style component improves relevance marginally while degrading coherence, consistency, and fluency more than it helps. We interpret this as evidence that a sharply localised candidate-to-source grounding (single best match, no penalty) is the strongest signal for SUMMEVAL on three of four dimensions, and that modifications which disperse this localisation hurt overall correlation.

The accepted component (the lead-bias prior) does *not* diffuse. It is a source-side structural prior that concentrates weight on high-information

Table B.12: LGS components tested during calibration. Development-split mean Spearman  $\rho$  averaged across the four SUMMEVAL dimensions on the 50-article development split. Only the lead-bias prior was retained in the canonical metric.

| Component                           | Form                                                                        | Dev $\rho$ at optimum           | Verdict                                                                         |
|-------------------------------------|-----------------------------------------------------------------------------|---------------------------------|---------------------------------------------------------------------------------|
| Recall baseline                     | $\text{mean}_j \max_i \cos(c_j, s_i)$                                       | .233                            | reference point.                                                                |
| Lead-bias prior $\lambda$           | $\text{mean}_j \max_i w(i) \cos(c_j, s_i),$<br>$w(i) = \exp(-\lambda i/n)$  | <b>.252</b> ( $\lambda^*=0.5$ ) | <b>accepted:</b> only component to lift the development mean over the baseline. |
| Legacy precision                    | $F_\beta$ $F_\beta(R, P)$ with $P$ over top-30% source by degree centrality | .228 ( $\beta=2$ , frac=.30)    | rejected: aids relevance only by $\sim+.02$ , loses 3/4 dimensions.             |
| Anchor coverage $C^\alpha$          | $R \cdot C^\alpha$ , $C =  \text{distinct argmax}_i /m$                     | .229 ( $\alpha=.25$ )           | rejected: optimum at $\alpha=0$ (no coverage).                                  |
| Within-summary diversity $D^\gamma$ | $R \cdot D^\gamma$ , $D = 1 - \text{mean}_{j < k} \cos(c_j, c_k)$           | .209 ( $\gamma=.10$ )           | rejected: monotonically harmful for $\gamma > 0$ .                              |
| Top- $k$ recall                     | $\text{mean}_j \frac{1}{k} \sum_{\text{top-}k} \cos(c_j, s_i)$              | .213 ( $k=2$ )                  | rejected: monotonically harmful for $k > 1$ .                                   |

positions *before* the argmax, without spreading the candidate-side aggregation. This is consistent with the diffusion hypothesis: priors that reshape the source without spreading the candidate-side aggregator are the operations most likely to help.

*Implication for future variants.* Future extensions of LGS proposed under the same protocol should be evaluated against this pattern. A new com-

ponent that multiplies recall by a candidate-side coverage or diversity term is unlikely to lift the development mean unless its formulation specifically avoids the diffusion problem documented here. Source-side priors (alternative decay shapes, salience-based weighting, structural weighting derived from discourse parsing) are the higher-prior direction for future work; we list specific candidates in Section 8.6.

## References

- [1] K. Papineni, S. Roukos, T. Ward, W.-J. Zhu, BLEU: a method for automatic evaluation of machine translation, in: Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics (ACL), 2002, pp. 311–318. doi:10.3115/1073083.1073135.  
URL <https://aclanthology.org/P02-1040/>
- [2] C.-Y. Lin, ROUGE: A package for automatic evaluation of summaries, in: Text Summarization Branches Out, Association for Computational Linguistics, Barcelona, Spain, 2004, pp. 74–81.  
URL <https://aclanthology.org/W04-1013/>
- [3] S. Banerjee, A. Lavie, METEOR: An automatic metric for MT evaluation with improved correlation with human judgments, in: Proceedings of the ACL Workshop on Intrinsic and Extrinsic Evaluation Measures for Machine Translation and/or Summarization, Association for Computational Linguistics, Ann Arbor, Michigan, 2005, pp. 65–72.  
URL <https://aclanthology.org/W05-0909/>

- [4] M. Popović, chrF: character n-gram F-score for automatic MT evaluation, in: Proceedings of the Tenth Workshop on Statistical Machine Translation (WMT), 2015, pp. 392–395. doi:10.18653/v1/W15-3049. URL <https://aclanthology.org/W15-3049/>
- [5] T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, Y. Artzi, BERTScore: Evaluating text generation with BERT, in: International Conference on Learning Representations (ICLR), 2020. URL <https://openreview.net/forum?id=SkeHuCVFDr>
- [6] W. Zhao, M. Peyrard, F. Liu, Y. Gao, C. M. Meyer, S. Eger, MoverScore: Text generation evaluating with contextualized embeddings and Earth Mover Distance, in: Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP), 2019, pp. 563–578. doi:10.18653/v1/D19-1053. URL <https://aclanthology.org/D19-1053/>
- [7] W. Yuan, G. Neubig, P. Liu, BARTScore: Evaluating generated text as text generation, in: Advances in Neural Information Processing Systems (NeurIPS), Vol. 34, 2021, pp. 27263–27277. URL <https://arxiv.org/abs/2106.11520>
- [8] R. K. Amplayo, P. J. Liu, Y. Zhao, S. Narayan, SMART: Sentences as basic units for text evaluation, arXiv preprint arXiv:2208.01030 (2022). doi:10.48550/arXiv.2208.01030. URL <https://arxiv.org/abs/2208.01030>

- [9] A. R. Fabbri, W. Kryściński, B. McCann, C. Xiong, R. Socher, D. Radev, SummEval: Re-evaluating summarization evaluation, *Transactions of the Association for Computational Linguistics* 9 (2021) 391–409. doi:10.1162/tacl\_a\_00373.  
URL <https://aclanthology.org/2021.tacl-1.24/>
- [10] M. Zhong, Y. Liu, D. Yin, Y. Mao, Y. Jiao, P. Liu, C. Zhu, H. Ji, J. Han, Towards a unified multi-dimensional evaluator for text generation, in: *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2022, pp. 2023–2038. doi:10.18653/v1/2022.emnlp-main.131.  
URL <https://aclanthology.org/2022.emnlp-main.131/>
- [11] Y. Liu, D. Iter, Y. Xu, S. Wang, R. Xu, C. Zhu, G-Eval: NLG evaluation using GPT-4 with better human alignment, in: *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023, pp. 2511–2522. doi:10.18653/v1/2023.emnlp-main.153.  
URL <https://aclanthology.org/2023.emnlp-main.153/>
- [12] T. Ito, K. van Deemter, J. Suzuki, Reference-free evaluation metrics for text generation: A survey, *arXiv preprint arXiv:2501.12011* (2025). doi:10.48550/arXiv.2501.12011.  
URL <https://arxiv.org/abs/2501.12011>
- [13] J. Gu, X. Jiang, Z. Shi, H. Tan, X. Zhai, C. Xu, W. Li, Y. Shen, S. Ma, H. Liu, S. Wang, K. Zhang, Y. Wang, W. Gao, L. Ni, J. Guo, A survey on LLM-as-a-judge, *arXiv preprint arXiv:2411.15594* (2024).

doi:10.48550/arXiv.2411.15594.

URL <https://arxiv.org/abs/2411.15594>

- [14] C. Kedzie, K. McKeown, H. Daumé III, Content selection in deep learning models of summarization, in: Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2018, pp. 1818–1828. doi:10.18653/v1/D18-1208.  
URL <https://aclanthology.org/D18-1208/>
- [15] M. Grusky, M. Naaman, Y. Artzi, Newsroom: A dataset of 1.3 million summaries with diverse extractive strategies, in: Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT), 2018, pp. 708–719. doi:10.18653/v1/N18-1065.  
URL <https://aclanthology.org/N18-1065/>
- [16] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, I. Polosukhin, Attention is all you need, in: Advances in Neural Information Processing Systems (NIPS), Vol. 30, 2017, pp. 5998–6008.  
URL <https://arxiv.org/abs/1706.03762>
- [17] J. Devlin, M.-W. Chang, K. Lee, K. Toutanova, BERT: Pre-training of deep bidirectional transformers for language understanding, in: Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT), 2019, pp. 4171–4186. doi:10.18653/v1/N19-1423.  
URL <https://aclanthology.org/N19-1423/>

- [18] M. Lewis, Y. Liu, N. Goyal, M. Ghazvininejad, A. Mohamed, O. Levy, V. Stoyanov, L. Zettlemoyer, BART: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension, in: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL), 2020, pp. 7871–7880. doi:10.18653/v1/2020.acl-main.703.  
URL <https://aclanthology.org/2020.acl-main.703/>
- [19] T. Sellam, D. Das, A. P. Parikh, BLEURT: Learning robust metrics for text generation, in: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL), 2020, pp. 7881–7892. doi:10.18653/v1/2020.acl-main.704.  
URL <https://aclanthology.org/2020.acl-main.704/>
- [20] R. Rei, C. Stewart, A. C. Farinha, A. Lavie, COMET: A neural framework for MT evaluation, in: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2020, pp. 2685–2702. doi:10.18653/v1/2020.emnlp-main.213.  
URL <https://aclanthology.org/2020.emnlp-main.213/>
- [21] W. Kryściński, B. McCann, C. Xiong, R. Socher, Evaluating the factual consistency of abstractive text summarization, in: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2020, pp. 9332–9346. doi:10.18653/v1/2020.emnlp-main.750.  
URL <https://aclanthology.org/2020.emnlp-main.750/>
- [22] P. Laban, T. Schnabel, P. N. Bennett, M. A. Hearst, SummaC: Re-



visiting NLI-based models for inconsistency detection in summarization,  
Transactions of the Association for Computational Linguistics 10 (2022)  
163–177. doi:10.1162/tac1\_a\_00453.

URL <https://aclanthology.org/2022.tac1-1.10/>

- [23] T. B. Brown, B. Mann, N. Ryder, M. Subbiah, J. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, S. Agarwal, A. Herbert-Voss, G. Krueger, T. Henighan, R. Child, A. Ramesh, D. M. Ziegler, J. Wu, C. Winter, C. Hesse, M. Chen, E. Sigler, M. Litwin, S. Gray, B. Chess, J. Clark, C. Berner, S. McCandlish, A. Radford, I. Sutskever, D. Amodei, Language models are few-shot learners, in: Advances in Neural Information Processing Systems (NeurIPS), Vol. 33, 2020, pp. 1877–1901.

URL <https://arxiv.org/abs/2005.14165>

- [24] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. H. Chi, Q. V. Le, D. Zhou, Chain-of-thought prompting elicits reasoning in large language models, in: Advances in Neural Information Processing Systems (NeurIPS), Vol. 35, 2022, pp. 24824–24837.

URL <https://arxiv.org/abs/2201.11903>

- [25] S. S. Balamurali, L. Cheng, Revisiting NLI: Towards cost-effective and human-aligned metrics for evaluating LLMs in question answering, arXiv preprint arXiv:2511.07659 (2025). doi:10.48550/arXiv.2511.07659.

URL <https://arxiv.org/abs/2511.07659>

- [26] C. Raffel, N. Shazeer, A. Roberts, K. Lee, S. Narang, M. Matena, Y. Zhou, W. Li, P. J. Liu, Exploring the limits of transfer learning

- with a unified text-to-text transformer, *Journal of Machine Learning Research* 21 (140) (2020) 1–67.  
URL <https://jmlr.org/papers/v21/20-074.html>
- [27] J. Fu, S.-K. Ng, Z. Jiang, P. Liu, GPTScore: Evaluate as you desire, arXiv preprint arXiv:2302.04166 (2023). doi:10.48550/arXiv.2302.04166.  
URL <https://arxiv.org/abs/2302.04166>
- [28] Y. Liu, M. Lapata, Text summarization with pretrained encoders, in: *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, 2019, pp. 3730–3740. doi:10.18653/v1/D19-1387.  
URL <https://aclanthology.org/D19-1387/>
- [29] J. Zhang, Y. Zhao, M. Saleh, P. J. Liu, PEGASUS: Pre-training with extracted gap-sentences for abstractive summarization, in: *Proceedings of the 37th International Conference on Machine Learning (ICML)*, 2020, pp. 11328–11339.  
URL <https://proceedings.mlr.press/v119/zhang20ae.html>
- [30] A. See, P. J. Liu, C. D. Manning, Get to the point: Summarization with pointer-generator networks, in: *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2017, pp. 1073–1083. doi:10.18653/v1/P17-1099.  
URL <https://aclanthology.org/P17-1099/>

- [31] N. Reimers, I. Gurevych, Sentence-BERT: Sentence embeddings using Siamese BERT-Networks, in: Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP), 2019, pp. 3982–3992. doi:10.18653/v1/D19-1410.  
URL <https://aclanthology.org/D19-1410/>
- [32] Z. Nussbaum, J. X. Morris, B. Duderstadt, A. Mulyar, Nomic embed: Training a reproducible long context text embedder, arXiv preprint arXiv:2402.01613 (2024). doi:10.48550/arXiv.2402.01613.  
URL <https://arxiv.org/abs/2402.01613>
- [33] K. M. Hermann, T. Kočiský, E. Grefenstette, L. Espeholt, W. Kay, M. Suleyman, P. Blunsom, Teaching machines to read and comprehend, in: Advances in Neural Information Processing Systems (NIPS), Vol. 28, 2015, pp. 1693–1701.  
URL <https://arxiv.org/abs/1506.03340>
- [34] J. Chen, S. Xiao, P. Zhang, K. Luo, D. Lian, Z. Liu, M3-Embedding: Multi-linguality, multi-functionality, multi-granularity text embeddings through self-knowledge distillation, in: Findings of the Association for Computational Linguistics: ACL 2024, 2024, pp. 2318–2335. doi:10.18653/v1/2024.findings-acl.137.  
URL <https://aclanthology.org/2024.findings-acl.137/>
- [35] B. Efron, R. J. Tibshirani, An Introduction to the Bootstrap, Vol. 57 of Monographs on Statistics and Applied Probability, Chapman & Hall/CRC, New York, 1993.

- [36] Y. Benjamini, Y. Hochberg, Controlling the false discovery rate: A practical and powerful approach to multiple testing, *Journal of the Royal Statistical Society: Series B (Methodological)* 57 (1) (1995) 289–300. doi:10.1111/j.2517-6161.1995.tb02031.x.
- [37] W. Wang, F. Wei, L. Dong, H. Bao, N. Yang, M. Zhou, MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers, in: *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 33, 2020, pp. 5776–5788.  
URL <https://arxiv.org/abs/2002.10957>
- [38] Qwen Team, Qwen2.5: A party of foundation models, <https://qwenlm.github.io/blog/qwen2.5/>, blog post, accessed 2026-05-31 (2024).
- [39] Ollama, Ollama: Get up and running with large language models, locally, <https://ollama.com>, software project page, accessed 2026-05-31 (2024).
- [40] T. Wolf, L. Debut, V. Sanh, J. Chaumond, C. Delangue, A. Moi, P. Cistac, T. Rault, R. Louf, M. Funtowicz, J. Davison, S. Shleifer, P. von Platen, C. Ma, Y. Jernite, J. Plu, C. Xu, T. Le Scao, S. Gugger, M. Drame, Q. Lhoest, A. M. Rush, HuggingFace’s Transformers: State-of-the-art natural language processing, in: *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, 2020, pp. 38–45. doi:10.18653/v1/2020.emnlp-demos.6.  
URL <https://aclanthology.org/2020.emnlp-demos.6/>

- [41] A. Cohan, F. Dernoncourt, D. S. Kim, T. Bui, S. Kim, W. Chang, N. Goharian, A discourse-aware attention model for abstractive summarization of long documents, in: Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT), 2018, pp. 615–621. doi:10.18653/v1/N18-2097.  
URL <https://aclanthology.org/N18-2097/>
- [42] B. Gliwa, I. Mochol, M. Biesek, A. Wawer, SAMSum corpus: A human-annotated dialogue dataset for abstractive summarization, in: Proceedings of the 2nd Workshop on New Frontiers in Summarization, 2019, pp. 70–79. doi:10.18653/v1/D19-5409.  
URL <https://aclanthology.org/D19-5409/>